



Corso di Laurea in Ingegneria Elettronica - Università degli Studi di Udine

Tecnologia e Progettazione di **MEMORIE NON VOLATILI**

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Tecnologia e Progettazione di **MEMORIE NON VOLATILI**



- PANORAMICA SULLE MEMORIE NON VOLATILI
- CELLA DI MEMORIA A FLOATING GATE
 - PRINCIPI DI FUNZIONAMENTO
 - COEFFICIENTI CAPACITIVI
 - SCRITTURA DELLA CELLA A FLOATING GATE
 - cancellazione UV
 - scrittura per Fowler-Nordheim tunnelling
 - programmazione per Channel Hot Electrons



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Tecnologia e Progettazione di **MEMORIE NON VOLATILI**

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- **DISPOSITIVO FLASH NOR**
 - **FUNZIONAMENTO DEL DISPOSITIVO**
 - organizzazione della matrice di memoria NOR
 - lettura, programmazione e cancellazione
 - **AFFIDABILITA' DELLA MEMORIA**
 - disturbi di programmazione e lettura
 - endurance e ritenzione
 - **TESTING E RESA**
 - **DISPOSITIVI MULTILIVELLO**
- **DISPOSITIVO FLASH NAND**
 - **FUNZIONAMENTO DEL DISPOSITIVO**
 - organizzazione della matrice di memoria NAND
 - lettura, programmazione e cancellazione



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Tecnologia e Progettazione di **MEMORIE NON VOLATILI**

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- **PROBLEMI DI SCALABILITA' DELLE MEMORIE FLASH**
 - Elementi attivi
 - Elementi passivi
- **ALTRE MEMORIE NON VOLATILI**
 - **FERAM**
 - **MRAM and STT-MRAM**
 - **RRAM**



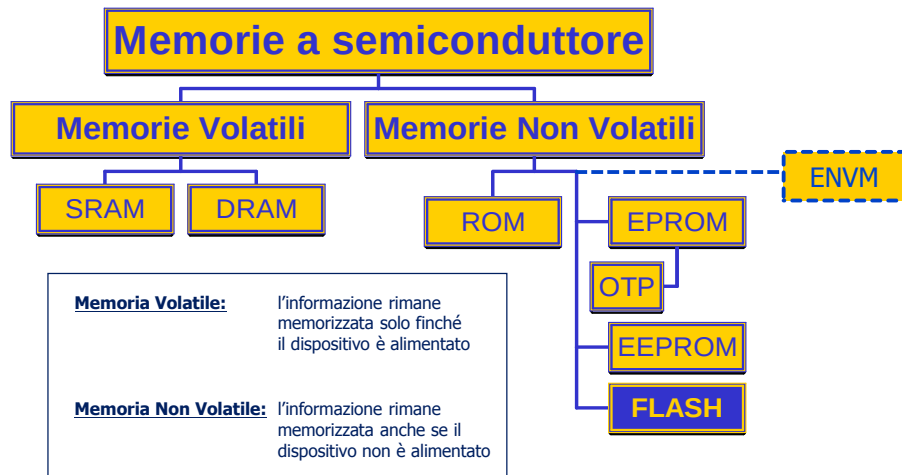
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- **AN OUTLOOK INTO THE FUTURE**
 - Una lezione sullo scaling
 - Memorie a cambiamento di fase (PCM)
 - Possibili evoluzioni delle memorie PCM
 - Memorie a cross-point

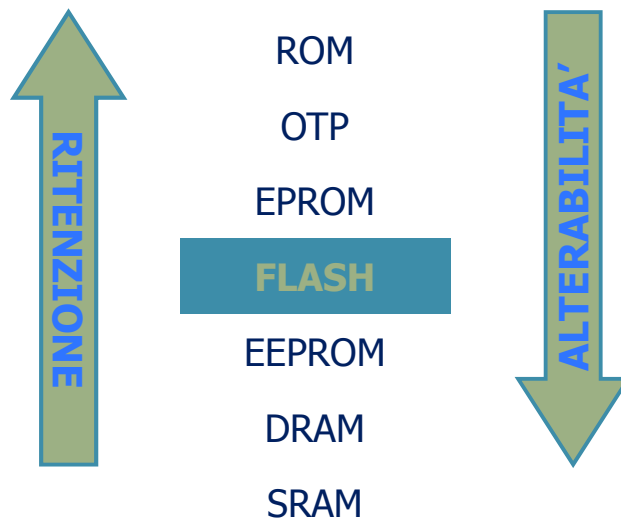
glossario

- **Array efficiency:** rapporto tra area della matrice di memoria e area dell'intero chip
- **Bit (binary digit):** unità base di memoria, "1" o "0"
- **BitLine:** linea di interconnessione per le operazioni di I/O della matrice di memoria
- **Byte:** gruppo di bits che vengono letti simultaneamente
- **Cancellazione:** l'operazione di rimozione di elettroni dalla floating gate
- **Cella:** il dispositivo a semiconduttore che immagazzina un bit (o più bit)
- **Control Gate:** gate di controllo del transistor di memoria tramite accoppiamento capacitivo
- **Disturbo:** indesiderato cambiamento dello stato della memoria durante le operazioni di scrittura o lettura
- **ECC (error correction code):** tecnica per correggere errori in una memoria migliorando affidabilità e resa
- **Endurance:** numero di cicli di scrittura/cancellazione che una memoria garantisce
- **Ferroelettrico:** materiale con caratteristiche di polarizzazione elettrica permanenti, la cui polarità può essere modificata mediante campo elettrico
- **Floating gate:** gate in silicio policristallino completamente isolata mediante dielettrici
- **Fowler-Nordheim tunneling:** tunnel di elettroni attraverso una barriera di potenziale triangolare, usato per la scrittura nelle EEPROM e nelle FLASH
- **Hot electron injection:** iniezione di elettroni caldi dal canale del transistor nella floating gate, oltre la barriera di potenziale dell'ossido
- **Memoria Non Volatile:** memoria i cui dati sono mantenuti senza la necessità di una alimentazione esterna
- **Memoria Volatile:** memoria i cui dati sono mantenuti mediante una alimentazione esterna e/o un continuo refresh
- **ONO (Ossido-Nitruro-Ossido):** dielettrico utilizzato come isolante tra control gate e floating gate
- **Ossido di tunnel:** ossido di gate abbastanza sottile da permettere il Fowler - Nordheim tunneling
- **Programmazione:** l'operazione di iniezione di elettroni nella floating gate
- **Ridondanza:** tecnica di progettazione che migliora la resa di un dispositivo mediante l'utilizzo di celle di scorta, che possono sostituire eventuali celle difettose
- **Ritenzione:** capacità di mantenere la carica immagazzinata nella cella
- **Scrittura:** l'operazione generica di variazione dello stato della memoria (programmazione o cancellazione)

Le memorie a semiconduttore



Memorie: proprietà fondamentali



Un esempio: l'iPod



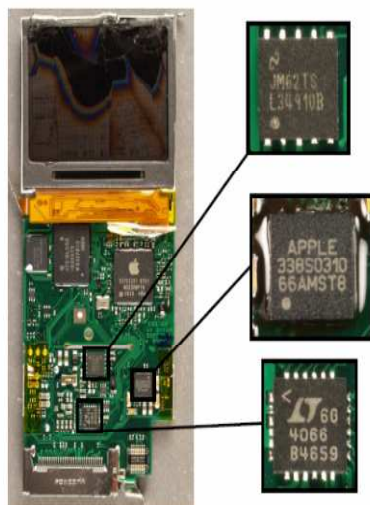
Cosa vediamo dall'esterno?

- **L'estetica**
- **L'interfaccia:**
 - Input:
 - Sensori tattili
 - Output
 - Display LCD
 - Cuffie



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Dentro l'iPod - 1



Step-down switching regulator

Audio CODEC

USB power manager



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Dentro l'iPod - 2



8Mb multi-purpose Flash NOR



256Mb mobile DRAM

ARM core DSP + Flash controller



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Dentro l'iPod - 3



32Gb (2x) multi-level NAND Flash



or
64Gb (dual-stacked)
multi-level NAND Flash

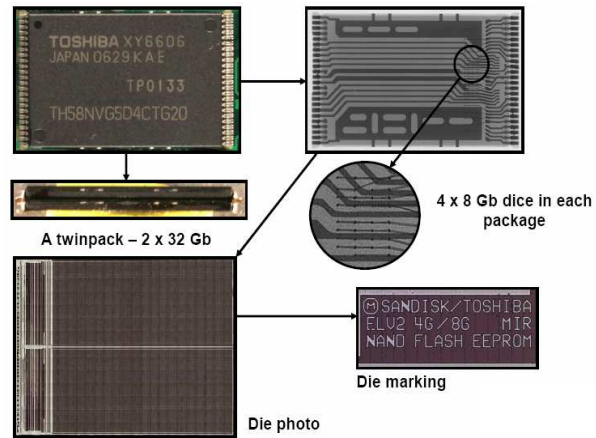


Power manager

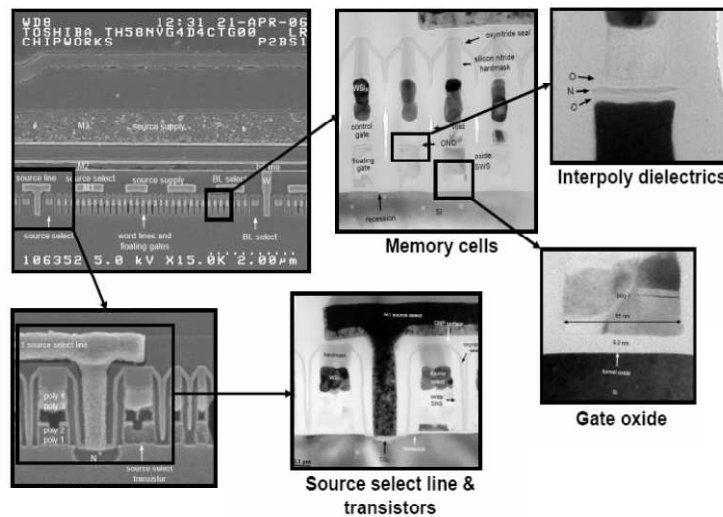


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32Gb (2x) NAND Flash



8Gb NAND dice

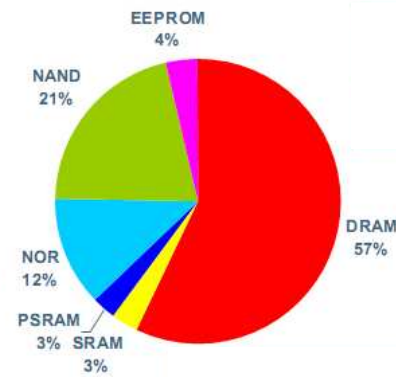


IC and memory markets

TOTAL IC MARKET
2006 Revenues \$259.7 billion
10.7% Growth



MEMORY IC MARKET
2006 Revenues \$60.3 billion
Market Percentage by Device



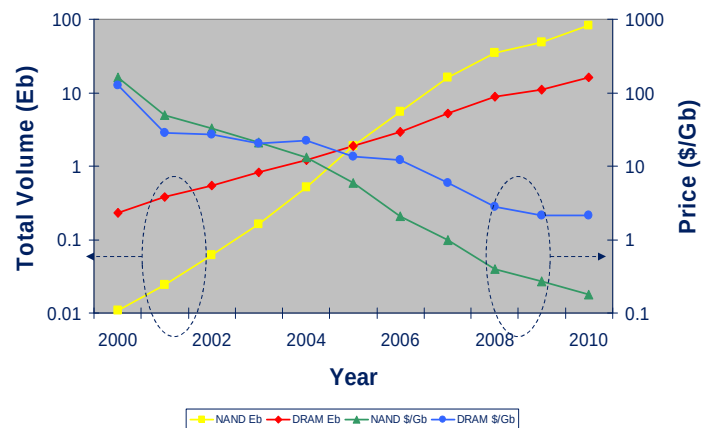
Source: Gartner Dataquest, 2007
Semiconductor Forecast Worldwide—Forecast Database [SEQS-WW-DB-DATA]



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Memory market



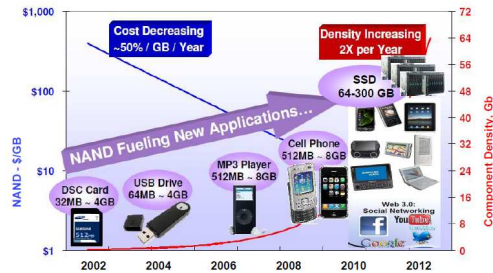
In the last years the production volume of DRAM and in particular of NAND Flash increased exponentially with a clear cost reduction trend



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NVM application boosting

The continuous decrease of the cost/Gb has boosted the introduction of NVM in a wide spectrum of applications



Dal transistor alla cella di memoria

soglia di un transistor
MOS n-channel

$$V_t = V_{FB} + V_S + 2|\Phi_p| + \frac{1}{C_{ox}} \sqrt{2\epsilon q N_a (2|\Phi_p| + V_S - V_B)}$$

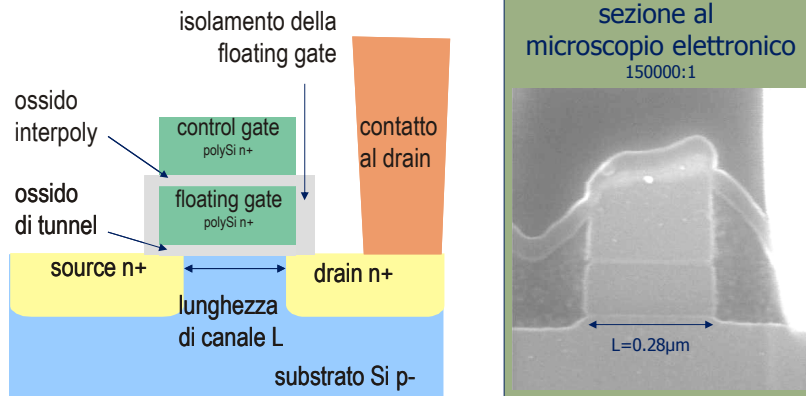
dipendenza della tensione
di *flat band* dalla carica
presente nell'ossido

$$V_{FB} = \Phi_{MS} - \frac{Q_f}{C_{ox}} - \frac{1}{C_{ox}} \int_0^{t_{ox}} x \rho(x) dx$$

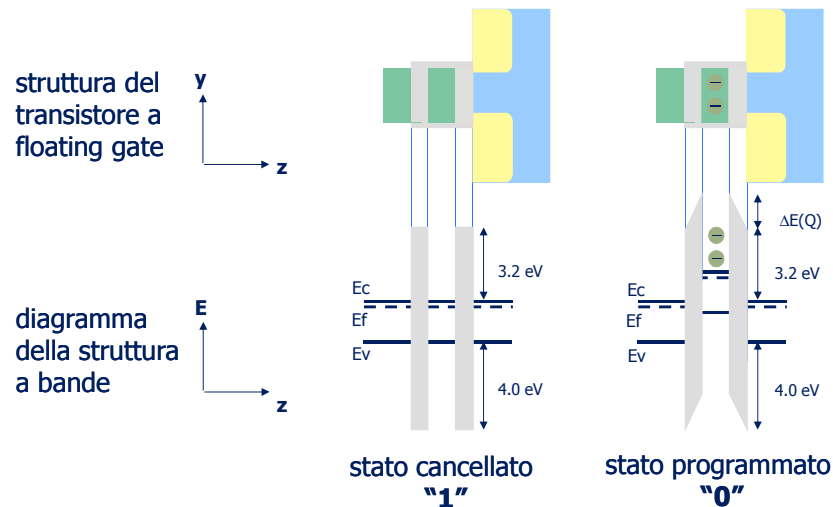
la tensione di soglia
di un transistor dipende
dalla carica presente
tra la gate e il substrato

$$V_t = V_t|_{Q=0} - k \cdot Q$$

La cella FLASH: sezione lungo L

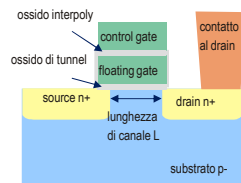
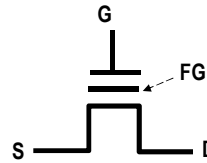
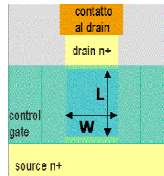


Struttura a bande della cella

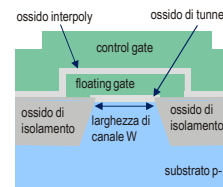


La cella FLASH a floating gate

layout cella Flash NOR



sezione lungo L

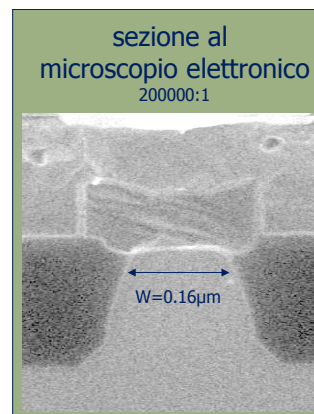
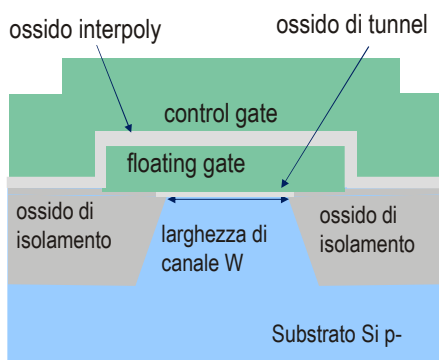


sezione lungo W



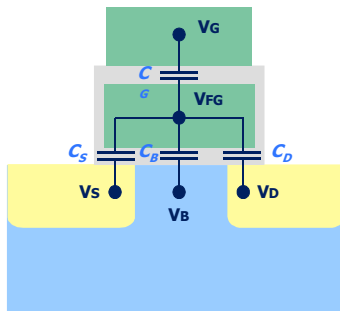
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La cella FLASH: sezione lungo W



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Floating gate e rapporti capacitivi



La cella di memoria a floating gate è schematizzabile come un circuito di 4 condensatori in parallelo:

$$\sum_{i=S,B,D,G} C_i \cdot (V_{FG} - V_i) = Q_{FG}$$

Definiti la capacità totale: $C_{TOT} = \sum_{i=S,B,D,G} C_i$

e i **rapporti capacitivi**: $\alpha_i = C_i / C_{TOT} < 1$

la tensione di floating gate dipende dalle tensioni ai capi della cella:

$$V_{FG} = \frac{Q_{FG}}{C_{TOT}} + \sum_{i=S,B,D,G} \alpha_i \cdot V_i$$

Corrente nella cella di memoria

Corrente in un transistor MOS:

$$I_D^{trans} = \beta^{trans} \cdot [(V_G^{trans} - V_T^{trans}) \cdot V_D]$$

Corrente in una cella a floating gate:

$$\begin{aligned} I_D^{cell} &= \beta^{trans} \cdot [(V_{FG} - V_T^{trans}) \cdot V_D] = \\ &= \beta^{trans} \cdot \left[\left(\alpha_G \cdot V_G^{cell} + \alpha_D \cdot V_D + \frac{Q_{FG}}{C_{TOT}} - V_T^{trans} \right) \cdot V_D \right] = \\ &= \alpha_G \cdot \beta^{trans} \cdot \left[\left(V_G^{cell} + \frac{Q_{FG}}{C_G} - \frac{V_T^{trans}}{\alpha_G} \right) \cdot V_D + \frac{\alpha_D}{\alpha_G} \cdot V_D^2 \right] \end{aligned}$$

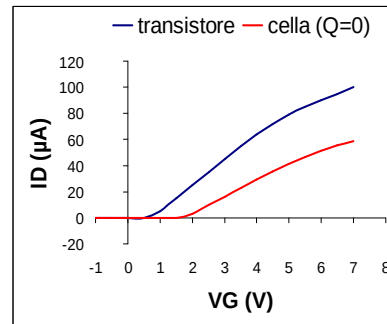
Soglia e guadagno della cella

Soglia della cella ($Q=0$):

$$V_T^{cell} \Big|_{(Q_{FG}=0)} = V_G^{cell} \Big|_{(I_D=0, V_D=0)} = \frac{V_T^{trans}}{\alpha_G}$$

Guadagno della cella:

$$\beta^{cell} = \frac{1}{V_D} \cdot \frac{\partial I_D^{cell}}{\partial V_G^{cell}} = \alpha_G \cdot \beta^{trans}$$



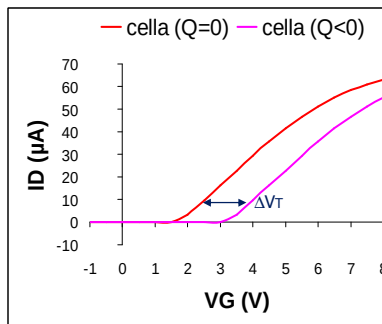
Carica nella floating gate

Soglia della cella:

$$V_T^{cell}(Q_{FG}) = \frac{V_T^{trans}}{\alpha_G} - \frac{Q_{FG}}{C_G} = V_T^{cell} \Big|_{(Q_{FG}=0)} - \frac{Q_{FG}}{C_G}$$



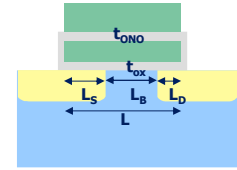
$$\Delta V_T^{cell}(Q_{FG}) = -\frac{Q_{FG}}{C_G}$$



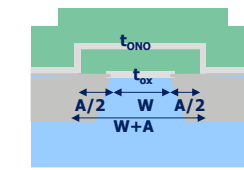
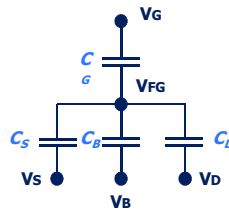
$$Q_{FG} = -C_G \cdot \Delta V_T^{cell}$$

$$V_{FG}(Q_{FG}) = \alpha_G \cdot (V_G^{cell} - \Delta V_T^{cell}) + \sum_{i=S,B,D} \alpha_i \cdot V_i$$

Calcolo dei rapporti capacitivi



sezione cella lungo L



sezione cella lungo W

In approssimazione di condensatori a piatti piani e paralleli:

$$C_G = \epsilon_{ONO} \cdot \epsilon_0 \cdot \frac{L \cdot (W + A)}{t_{ONO}} \Rightarrow \alpha_G = \frac{C_G}{C_{TOT}} = \frac{1}{1 + \frac{W}{W + A} \cdot \frac{t_{ONO}}{t_{ox}}}$$

$$C_{S,B,D} = \epsilon_{ox} \cdot \epsilon_0 \cdot \frac{L_{S,B,D} \cdot W}{t_{ox}} \Rightarrow \alpha_{S,B,D} = \frac{C_{S,B,D}}{C_{TOT}} = \frac{\frac{L_{S,B,D}}{L}}{1 + \frac{W + A}{W} \cdot \frac{t_{ox}}{t_{ONO}}}$$

	α_G	α_D
L	-	-
L _D	-	-
W	-	-
A	-	-
t _{ONO}	-	-
t _{ox}	-	-



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La cella come elettrometro

La cella di memoria a floating gate è un elettrometro ad elevatissima risoluzione: dati i parametri costruttivi di una cella in tecnologia 0.18µm

L	W	A	t _{ONO}	
0.3µm	0.2µm	0.25µm	15nm	$C_G = \epsilon_{ONO} \cdot \epsilon_0 \cdot \frac{L \cdot (W + A)}{t_{ONO}} \cong 0.3 \text{ fF}$

la variazione di carica pari a 1 elettrone (!) provoca una variazione di soglia della cella pari a:

$$\Delta V_T^{cell}(1e) = -\frac{Q_{FG}}{C_G} = \frac{1.6 \cdot 10^{-19} \text{ C}}{0.31 \text{ fF}} \cong 0.5 \text{ mV}$$

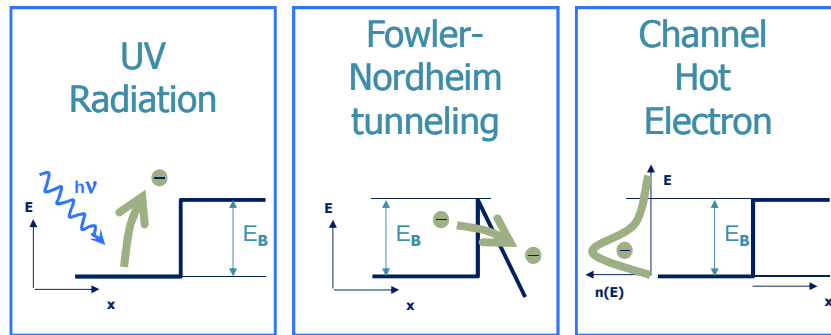
Variazioni di carica pari a poche decine di elettroni nella cella di memoria sono misurabili e possono avere un impatto significativo sui dispositivi con memorie non volatili



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Meccanismi di scrittura della cella

I meccanismi fisici utilizzati per la scrittura* di una cella di memoria a floating gate sono i seguenti:



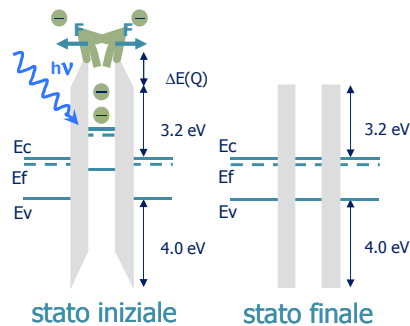
*si definisce scrittura l'operazione generica di variazione dello stato della cella di memoria



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Cancellazione per UV radiation

- Per effetto fotoelettrico si fornisce ai portatori della floating gate energia sufficiente a superare la barriera di potenziale dell'ossido
 - L'eventuale carica intrappolata nella floating gate genera il campo elettrico interno che causa la corrente netta con la quale il sistema si porta all'equilibrio (potenziale elettrochimico costante, quindi nessuna carica nella floating gate)
- L'irraggiamento UV è una operazione di reset
 - Non è detto che alla condizione di neutralità corrisponda uno stato logico della memoria, ma l'operazione è detta cancellazione poiché non è selettiva



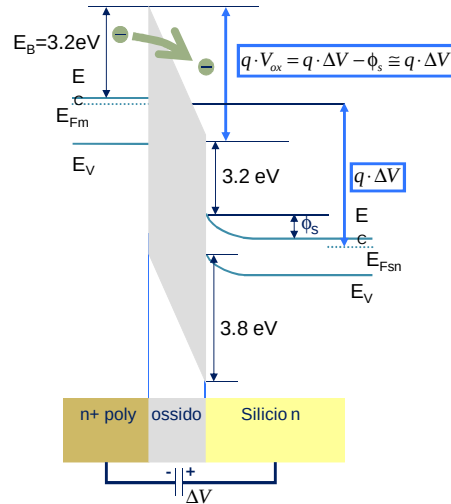
$$h\nu > E_B = 3.2 \text{ eV} \Rightarrow \begin{cases} \nu > 7.7 \times 10^{14} \text{ Hz} \\ \lambda < 380 \text{ nm} \end{cases}$$



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Fowler-Nordheim Tunneling

- Applicando una differenza di potenziale ΔV sufficientemente elevata tra le armature di un condensatore MOS e' possibile piegare le bande in modo tale da ottenere nell'ossido una **barriera di potenziale triangolare** attraverso la quale la probabilita' di tunnelling e' diversa da zero.
- Si ottiene cosi' una corrente di gate con la quale e' possibile alterare lo stato di carica di una gate flottante e quindi scrivere una memoria non volatile (il meccanismo e' bidirezionale).
- Per limitare la caduta di potenziale ϕ_s sul silicio, la giunzione n* che costituisce uno dei due capi del condensatore deve essere molto drogata.



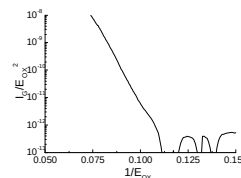
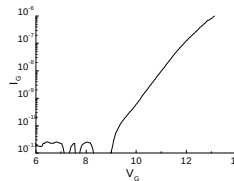
Equazione di Fowler-Nordheim

Il tunneling attraverso una barriera di potenziale triangolare può essere calcolato mediante il metodo WKB:

$$|J_{FN}| = A_{FN} \cdot F_{ox}^2 \cdot \exp\left(-\frac{B_{FN}}{F_{ox}}\right)$$

dove

$$\begin{cases} A_{FN} = \frac{e^3}{8\pi h} \cdot \frac{m_0}{m_{ox}^{eff}} \cdot \frac{1}{E_B} \\ B_{FN} = \frac{8\pi}{3eh} \cdot \sqrt{2 \cdot m_{ox}^{eff}} \cdot E_B^3 \end{cases}$$



Cancellazione per FN tunneling

Il meccanismo di Fowler-Nordheim tunneling viene utilizzato per cancellare una memoria Flash NOR, cioè per togliere elettroni dalla floating gate.

Analisi della cancellazione FN a tensioni esterne costanti:

Campo elettrico

$$F_{ox} = \frac{V_s - V_{FG}}{t_{ox}} = \frac{(1 - \alpha_s) \cdot V_s - \alpha_G \cdot V_G + \alpha_G \cdot \Delta V_T^{cell}}{t_{ox}}$$

Corrente di tunneling FN

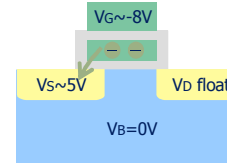
$$I_{FN} = S_{SG} \cdot J_{FN} = - \frac{dQ_{FG}}{dt} = C_G \cdot \frac{d\Delta V_T^{cell}}{dt} = C_{TOT} \cdot t_{ox} \cdot \frac{dF_{ox}}{dt}$$

Equazione differenziale della cancellazione per tunneling FN

$$\frac{dF_{ox}}{dt} = \frac{S_{SG} \cdot J_{FN}}{C_{TOT} \cdot t_{ox}} = - \frac{S_{SG} \cdot A_{FN}}{C_{TOT} \cdot t_{ox}} \cdot F_{ox}^2 \cdot \exp\left(-\frac{B_{FN}}{F_{ox}}\right)$$

Soluzione

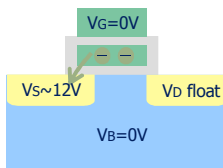
posti $\begin{cases} \frac{t_{ox} \cdot B_{FN}}{\alpha_G} = V_0 ; & \frac{S_{SG} \cdot A_{FN} \cdot B_{FN}}{C_{TOT} \cdot t_{ox}} = \frac{1}{t_0} \\ \frac{1 - \alpha_s}{\alpha_G} \cdot V_s - V_G = V^* \end{cases} \Rightarrow \Delta V_T^{cell}(t) = \frac{V_0}{\ln\left[\exp\left(\frac{V_0}{V^* + \Delta V_T^{cell}(0)}\right) + \frac{t}{t_0}\right]} - V^*$



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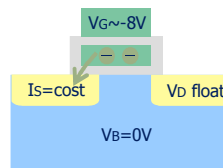
Cancellazione FN di una Flash NOR

I generazione



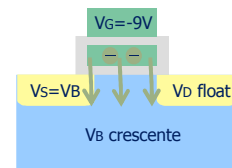
- + tensioni positive (semplicità circuitale)
- tensione V_{as} elevata (giunzione source graduale)
- correnti BBT elevate (alimentazione esterna)
- campo elettrico sull'ossido decrescente nel tempo

II generazione



- + correnti BBT ridotte (alimentazione interna)
- + campo elettrico sull'ossido costante nel tempo
- tensione V_{as} elevata (giunzione source graduale)
- tensioni negative (complessità circuitale)

III generazione



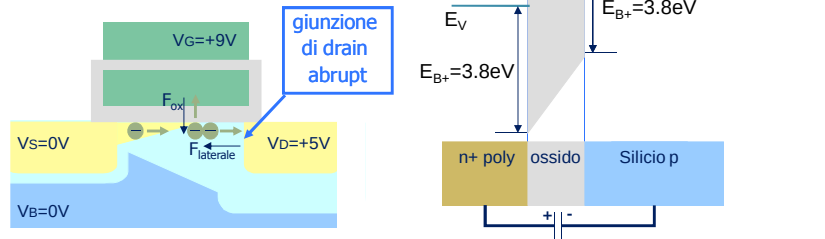
- + correnti BBT nulle (alimentazione interna)
- + tensione V_{as} nulla (giunzione source abrupt)
- + campo elettrico sull'ossido costante nel tempo
- tensioni negative (complessità circuitale)
- polarizzazione substrato (processo "triplo well")



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Channel Hot Electrons

- Gli elettroni che viaggiano dal source al drain guadagnano energia per effetto del campo elettrico laterale e la perdono per interazioni con il cristallo
- La distribuzione di energia degli elettroni $n(E)$ presenta una coda di elettroni con energia molto elevata (**Channel Hot Electrons**), superiore alla barriera di potenziale dell'ossido, nei punti del canale dove il campo elettrico laterale è molto elevato



La cella come amperometro

La cella di memoria a floating gate può essere utilizzata come un amperometro molto sensibile: dati i parametri costruttivi di una cella in tecnologia 0.18μm

L	W	A	t _{ONO}
0.3μm	0.2μm	0.25μm	15nm



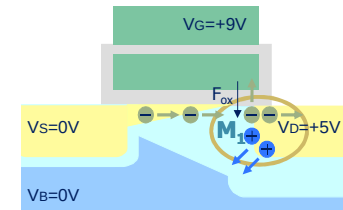
$$C_G = \epsilon_{ONO} \cdot \epsilon_0 \cdot \frac{L \cdot (W + A)}{t_{ONO}} \cong 0.3 \text{ fF}$$

le curve di programmazione effettuate con un impulsatore standard permettono di rilevare indirettamente correnti di gate estremamente basse:

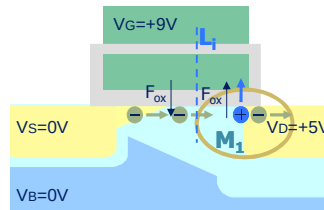
$$I_G = \frac{dQ_{FG}}{dt} = -C_G \cdot \frac{d\Delta V_T^{cell}}{dt}$$

La variazione di soglia di 1 V in 1s (programmazione a bassi campi) corrisponde su questa cella ad una corrente media di gate pari a 0.3fA.

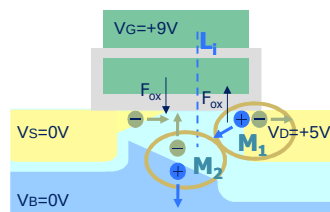
Moltiplicazione di elettroni e lacune



Iniezione di **Hot Electrons** prodotti da moltiplicazione per ionizzazione da impatto al drain



Iniezione di **Hot Holes** prodotti da moltiplicazione per ionizzazione da impatto al drain

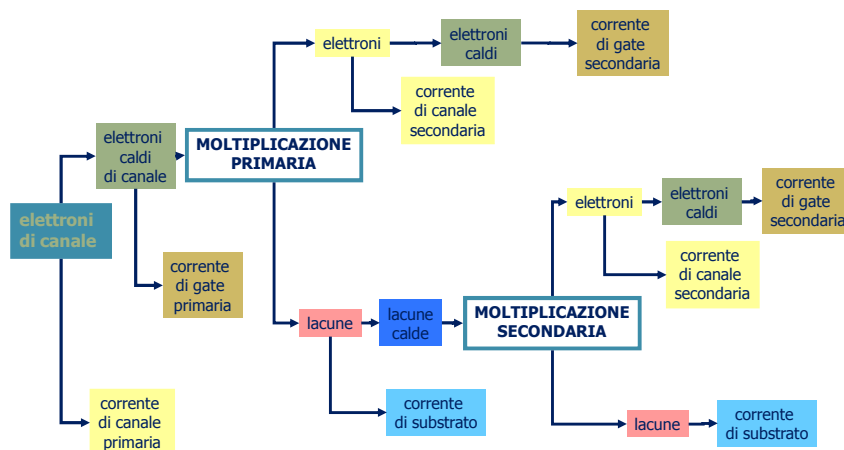


Iniezione di **Hot Electrons secondari** prodotti da moltiplicazione per ionizzazione secondaria da impatto nel substrato



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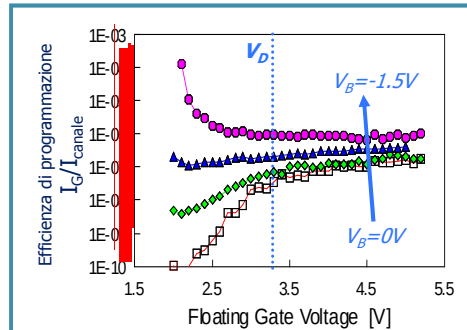
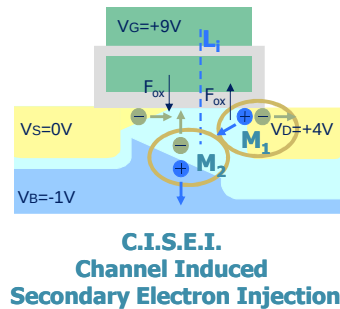
Iniezione di carica: moltiplicazione



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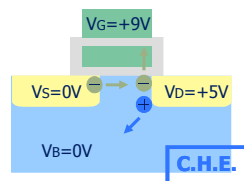
C.I.S.E.I.: programmazione con body

Quando l'efficienza di programmazione diventa bassa a causa del campo elettrico sfavorevole nella regione di drain ($V_{FG} < V_D$), un significativo incremento della corrente di gate si può ottenere polarizzando il substrato (operazione che richiede un processo con "triplo well");
l'aumento di efficienza di programmazione permette di ridurre la corrente assorbita dalla cella



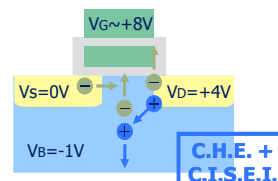
Programmazione di Flash NOR

I generazione



- + tensioni positive (semplicità circuitale)
- correnti di programmazione elevate

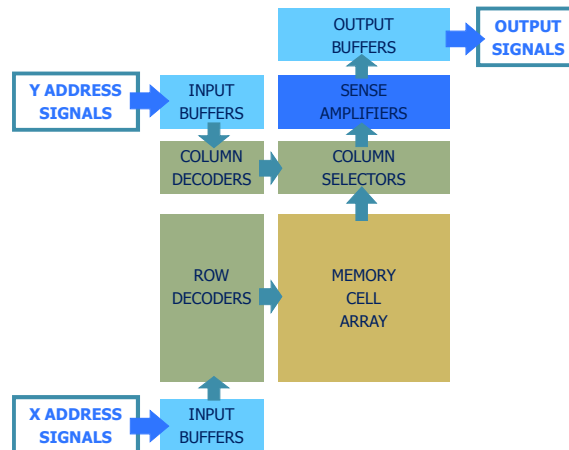
II generazione



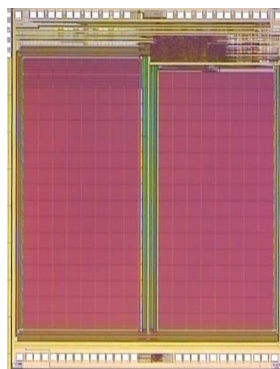
- + correnti di programmazione ridotte (alto parallelismo)
- tensioni negative (complessità circuitale)
- polarizzazione substrato (triplo well)

Memoria Flash NOR: organizzazione

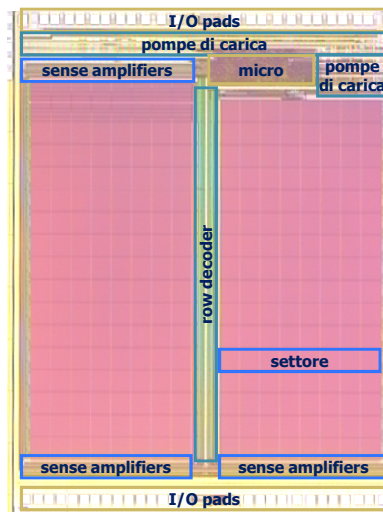
- La funzione di un dispositivo di memoria è quella di conservare informazioni e renderle disponibili in modo ordinato.
- In una memoria Flash a singolo livello i dati sono immagazzinati in forma digitale in celle di memoria, disposte secondo un arrangiamento a matrice.
- La capacità in bit della memoria è pari al numero di celle di memoria disponibili



Layout di un dispositivo Flash

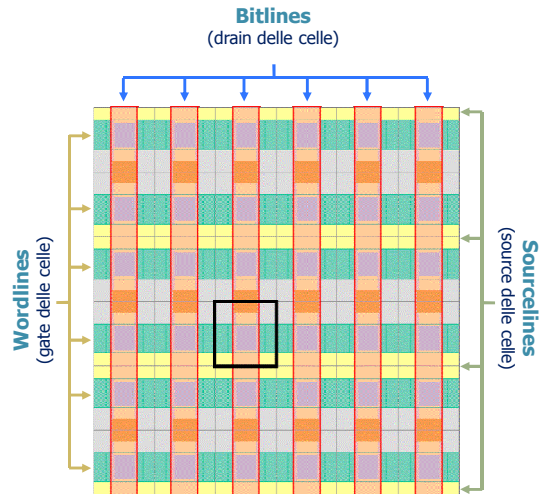
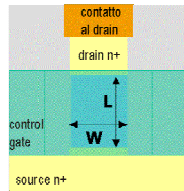


16Mbit Flash 3.0V
Tecnologia 0.25μm
chip size=28 mm²



Matrice di memoria Flash NOR

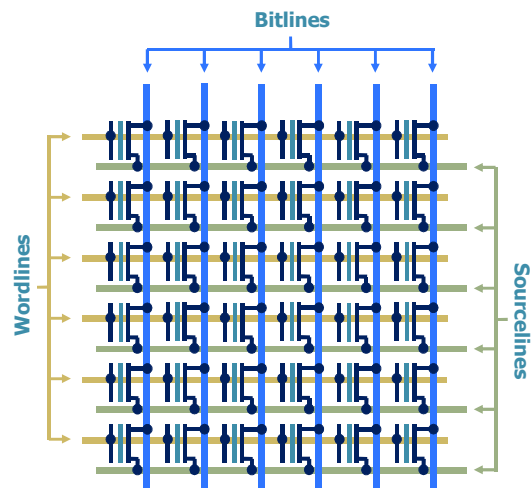
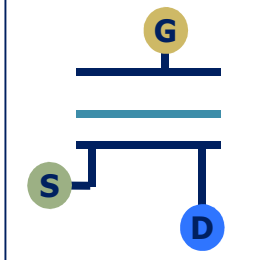
Cella Flash NOR singola



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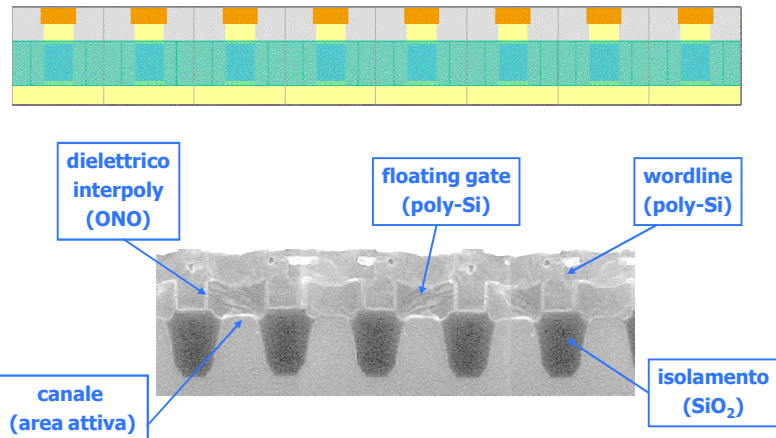
Circuito della matrice Flash NOR

Cella Flash NOR singola



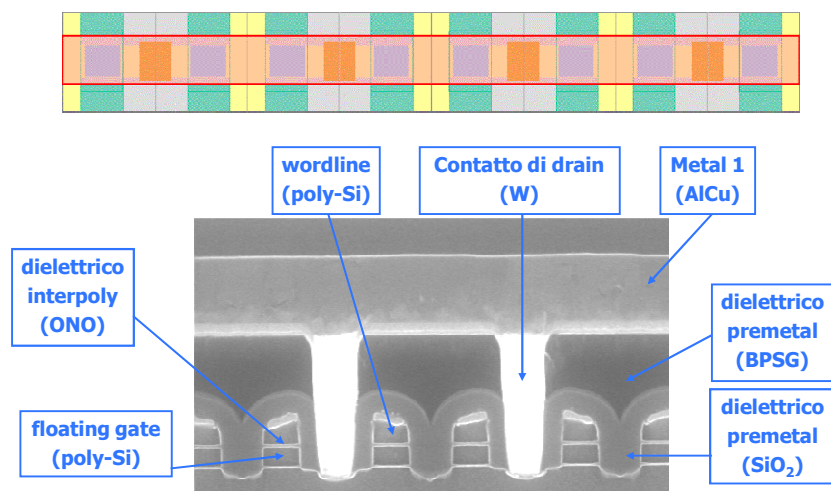
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Wordline di una matrice Flash



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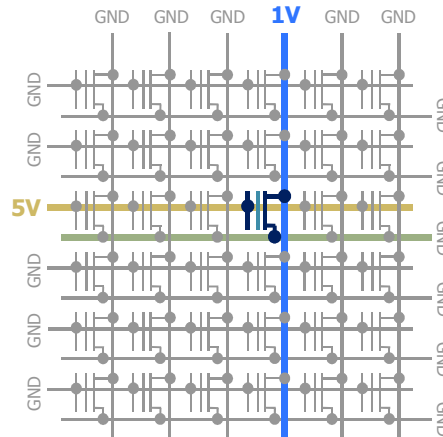
Bitline di una matrice Flash



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Memoria Flash: lettura

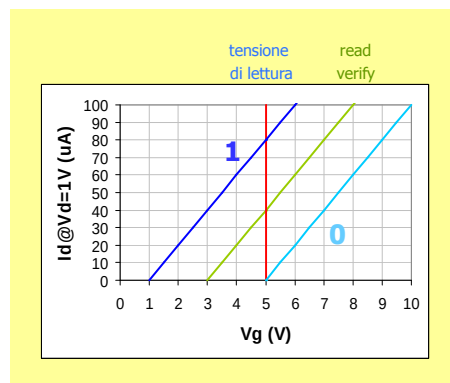
- L'organizzazione a matrice del dispositivo Flash NOR permette di selezionare e **leggere una singola cella**
- Per leggere una cella si alza la sua **wordline a ~5V** e la sua **bitline a ~1V**, con source e body a massa



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Lettura: read verify

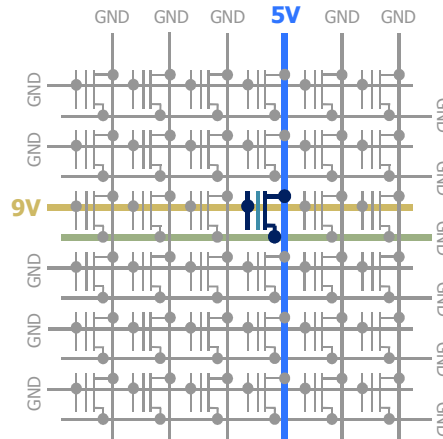
- In base al funzionamento del **circuito di sensing**, si definiscono **1** le celle che portano **più corrente** della cella di **read verify** alla tensione di lettura
- Si definiscono **0** le celle che portano **meno corrente** del **read verify** alla tensione di lettura
- Tempi di accesso
 - Random: 50 ÷ 150ns
 - Burst mode: 15 ÷ 30ns



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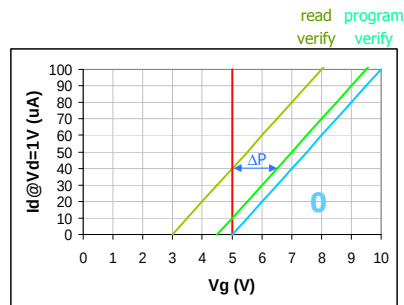
Memoria Flash: programmazione

- L'organizzazione a matrice del dispositivo Flash NOR permette di selezionare e **programmare una singola cella** per C.H.E.
- Per programmare una cella si alza la sua **wordline a ~9V** e la sua **bitline a ~5V**, con source e body a massa



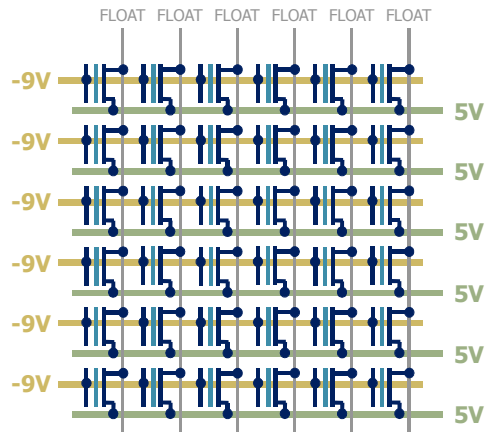
Programmazione: program verify

- Ogni impulso di programmazione (pochi μs) è seguito da una verifica della cella rispetto alla cella di **program verify**
- Tempi di program: $\sim 10 \mu s$
- Una cella è programmata quando **meno corrente** del **program verify** alla tensione di lettura
- Il **margin** di **programmazione ΔP** serve per l'affidabilità della cella (ritenzione e disturbi)



Memoria Flash: cancellazione

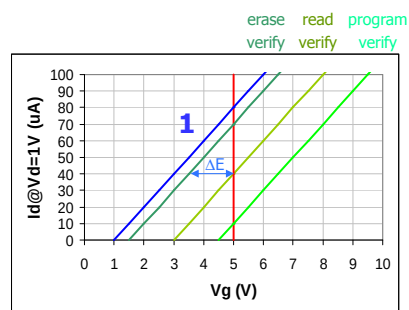
- L'organizzazione a matrice del dispositivo Flash NOR **non permette di cancellare una singola cella**
- La cancellazione viene eseguita per Fowler Nordheim tunneling su un intero blocco di celle (**settore**) portando tutte le **wordlines a ~-9V** e tutte le **sourcelines a ~5V**, con body a massa e drain floating



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Cancellazione: erase verify

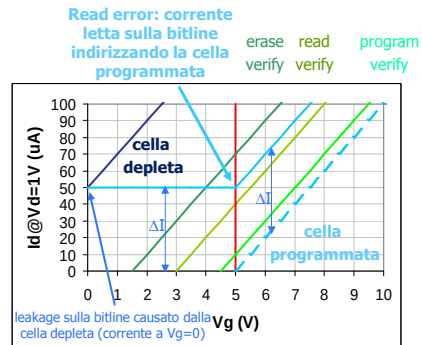
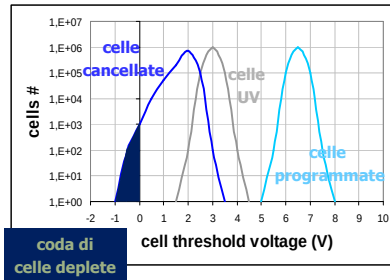
- Ogni impulso di cancellazione (decine di ms) è seguito da una verifica della cella rispetto alla cella di **erase verify**
- Una cella è cancellata quando **più corrente** dell'**erase verify** alla tensione di lettura
- Il **margin** di cancellazione ΔE serve per l'affidabilità della cella nel tempo (ritenzione e disturbi)



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Celle deplete

La cancellazione per FN è funzione esponenziale del campo elettrico sull'ossido di tunnel: piccole differenze di campo (indotte da cariche nell'ossido o dispersione di processo) causano sensibili differenze nella velocità di cancellazione. Cancellando un settore Flash è normale ottenere una certa percentuale di **celle deplete**.

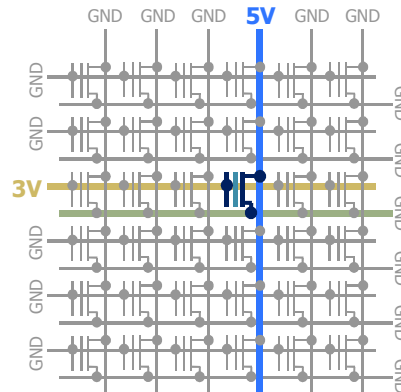
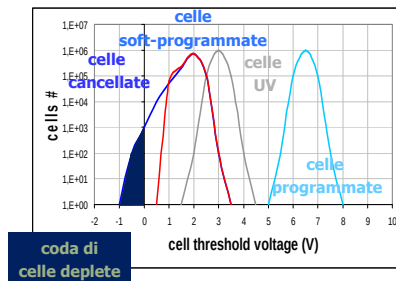


La presenza di celle deplete su una bitline può causare errori nella lettura delle altre celle della bitline, poiché la cella depleta aggiunge un offset di corrente ΔI sulla bitline



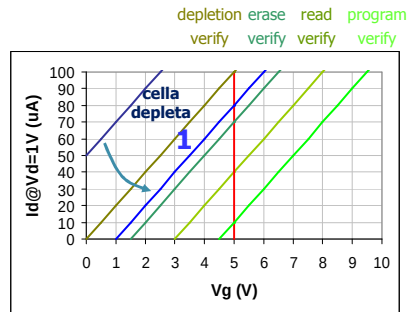
Soft-program delle celle deplete

Per recuperare le celle deplete si ricorre alla **soft-programmazione**: le celle a bassa soglia vengono programmate selettivamente per C.H.E. con tensione di gate molto bassa, in modo da evitare che la loro soglia finale superi il valore di erase verify

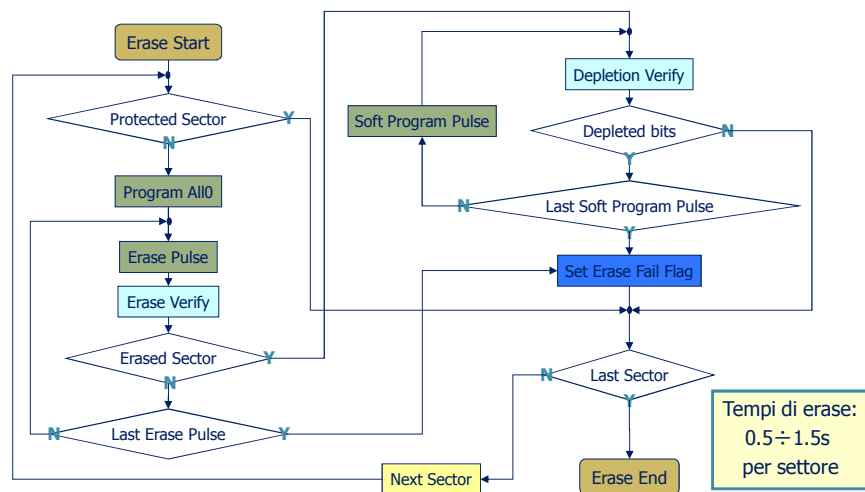


Cancellazione: depletion verify

- Ogni impulso di softprogrammazione (pochi μs) è seguito da una verifica della cella rispetto alla cella di **depletion verify**
- Una cella è softprogrammata quando **più corrente** del **depletion verify** alla tensione di lettura, ma porta comunque **meno corrente** dell' **erase verify**



Flash NOR: sequenza di cancellazione



Affidabilità di una memoria Flash

- Disturbi
 - disturbi in programmazione, disturbi in lettura
- Fast erasing bits
- Endurance
 - Degrado in ciclatura, bit erratici
- Ritenzione
 - leakage negli ossidi, contaminazione ionica, SILC



Disturbi di programmazione

Durante la programmazione della cella **A**, la cella **B** che condivide la stessa bitline e la cella **C** che sta sulla stessa wordline subiscono degli stress

B programmata:

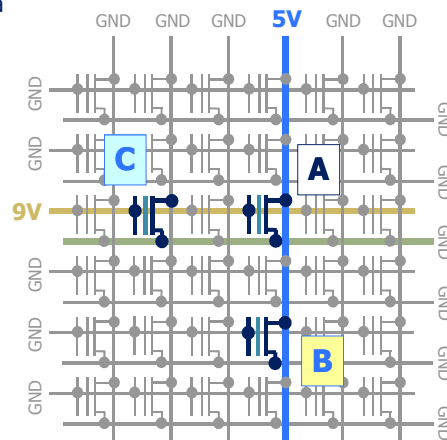
- FN drain stress su ossido di tunnel
- Hot Holes Injection
- $T = \Sigma$ program celle della bitline

C programmata:

- FN gate stress su ossido interpoly
- $T = \Sigma$ program celle della wordline

C cancellata:

- FN gate stress su ossido di tunnel
- $T = \Sigma$ program celle della wordline

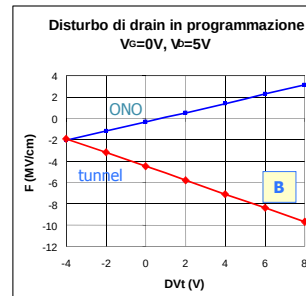
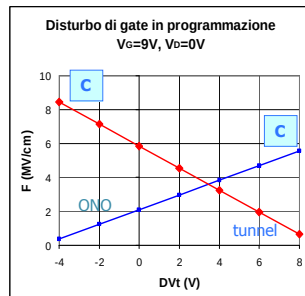


Disturbi: campi elettrici

I campi elettrici sugli ossidi attivi di una cella dipendono dalle tensioni applicate e dalla soglia della cella.

$$F_{ONO} = \frac{V_G - V_{FG}}{t_{ONO}} = \frac{(1 - \alpha_G) \cdot V_G + \alpha_G \cdot \Delta V_T - \alpha_D \cdot V_D}{t_{ONO}}$$

$$F_{ox} = \frac{V_{FG} - V_D}{t_{ox}} = \frac{\alpha_G \cdot (V_G - \Delta V_T) - (1 - \alpha_D) \cdot V_D}{t_{ox}}$$



Disturbi di lettura

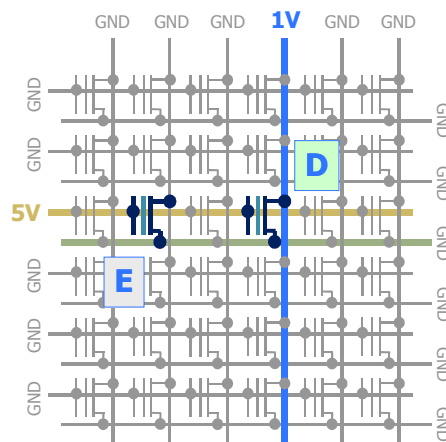
Durante la lettura della cella **D** essa tende a programarsi, mentre la cella **E** che condivide la stessa wordline subisce uno stress di gate

D cancellata:

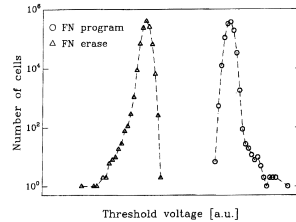
- Channel Hot Electrons Injection
- T= lifetime del dispositivo

E cancellata:

- FN gate stress su ossido di tunnel
- T= lifetime del dispositivo



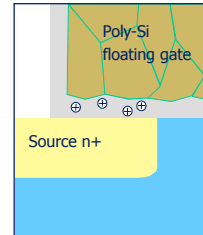
Fast erasing bits



Durante le operazioni di scrittura per Fowler-Nordheim tunneling si osservano normalmente code nella distribuzione delle celle (celle che si cancellano più velocemente della media)

Modelli fisici che spiegano l'aumento locale di campo elettrico:

- struttura a grani del poly
- cariche positive nell'ossido di tunnel



I bit più veloci vengono normalmente scartati poichè sono potenziali difetti in ciclatura



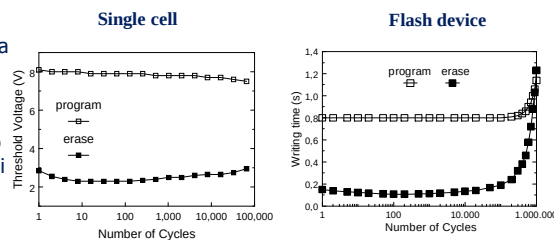
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Endurance di celle Flash

Le Memorie Non Volatili hanno una specifica di endurance variabile tra 100 cicli (EPROM) e 10^6 cicli (EEPROM).

Per una Flash la specifica tipica è di 10^5 cicli di scrittura e cancellazione.

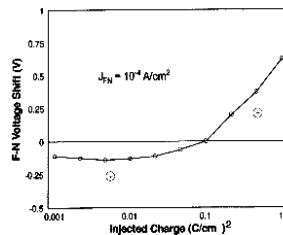
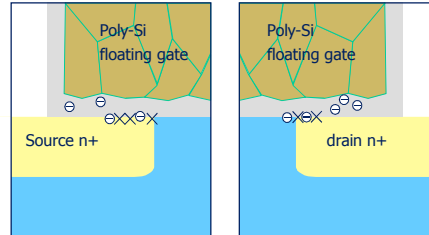
Durante la ciclatura di una cella Flash si osserva tipicamente la chiusura della finestra di funzionamento, cioè il contemporaneo degrado delle prestazioni (salto di soglia della cella) in programmazione C.H.E. e in cancellazione FN



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Degrado in ciclatura

Il degrado delle prestazioni della cella Flash che si osserva in ciclatura è dovuto al progressivo aumento di cariche negative nell'ossido di tunnel e all'interfaccia ossido-silicio, sia al source che al drain.



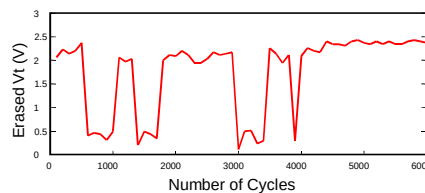
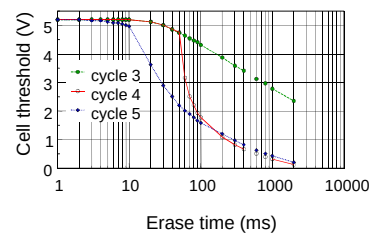
Si osserva infatti che con il continuo passaggio di cariche negative nell'ossido la tensione necessaria a sostenere un certo flusso di corrente aumenta (FN Voltage Shift)



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Bit erratici

- La fluttuazione delle cariche positive nell'ossido può causare il fenomeno dei **bit erratici**
- Un bit erratico si comporta in maniera imprevedibile durante la ciclatura
- Il controllo della qualità dell'ossido di tunnel e la riduzione dell'iniezione di cariche positive permette di limitare il fenomeno



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Ritenzione di carica

Una cella Flash deve garantire il suo stato di carica per 10 anni.

Data una cella con i seguenti parametri tecnologici:

L	W	A	t _{ONO}
0.3μm	0.2μm	0.25μm	15nm

$$C_G = \epsilon_{ONO} \cdot \epsilon_0 \cdot \frac{L \cdot (W + A)}{t_{ONO}} \cong 0.3 \text{ fF}$$

Ipotizzando una cella con margini di programmazione e cancellazione pari a 1 V, la carica massima che può essere persa dalla floating gate in 10 anni è:

$$Q_{FG} = -C_G \cdot \Delta V_T^{cell} = -0.3 \text{ fF} \cdot 1 \text{ V} = -0.3 \text{ fC} \cong 2000 \text{ elettroni}$$

Questa perdita di carica equivale ad un leakage medio di

$$I_{leakage} = \frac{2000 \text{ elettroni}}{10 \text{ anni}} \cong 10^{-24} \text{ A}$$



$$J_{leakage} = \frac{10^{-24} \text{ A}}{0.3 \mu\text{m} \cdot 0.2 \mu\text{m}} \cong 10^{-15} \text{ A/cm}^2$$

Ci sono due meccanismi di perdita di carica:

leakage attraverso gli ossidi e **contaminazione ionica**



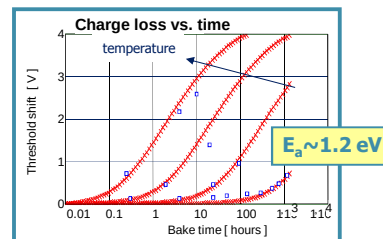
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Ritenzione: leakage sugli ossidi

Perdita di carica intrinseca

tutte le celle sono soggette a perdita di carica attraverso gli ossidi attivi, per FN tunneling o conduzione attraverso trappole nell'ossido

$$I_{leakage} \propto \exp\left(-\frac{E_a}{kT}\right)$$



Perdita di carica su single bit

difetti negli ossidi possono variare la ritenzione delle celle

ossido	meccanismo di leakage	dipendenza dal campo elettrico	energia di attivazione
tunnel	FN tunneling attraverso barriera di potenziale ridotta	esponenziale	$E_a \sim 0.3 \text{ eV}$
	conduzione per impurezze o difetti	lineare	$E_a \sim 0.6 \text{ eV}$
ONO	conduzione per emissione da trappole nel dielettrico	esponenziale	$E_a > 0.8 \text{ eV}$



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Ritenzione: contaminazione ionica

- La contaminazione ionica è causata dalla presenza di cariche mobili nei dielettrici del dispositivo
 - Sostanze introdotte durante la fabbricazione e non rimosse
 - Sostanze che entrano nel circuito durante la vita del dispositivo per la scarsa efficacia della passivazione
- La contaminazione ionica provoca problemi di ritenzione poichè le cariche mobili influenzano elettrostaticamente il potenziale della floating gate

Ritenzione: SILC

- per cancellare una Flash con un tunnel oxide da 10 nm in 100 ms bisogna applicare uno stress di $1e-4$ A/cm² a una tensione di 10 V
- dopo l'applicazione dello stress, la corrente di leakage aumenta (**Stress Induced Leakage Current**)
- Il leakage anomalo nell'ossido di tunnel avviene attraverso le trappole create dal passaggio di carica, quindi dopo ciclatura
- con la tecnologia attuale, per garantire la ritenzione dopo ciclatura il minimo spessore dell'ossido di tunnel è 8 nm

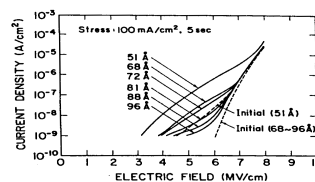


Fig. 1 J-E characteristics measured by capacitors having 51-96 Å oxide thickness before and after charge injection stress.

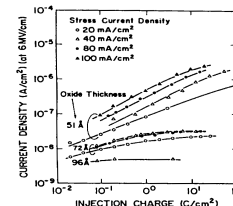


Fig. 3 Stress induced current at 6MV/cm as a function of injection charge passed through the capacitor oxide (51, 72, 96 Å).

Testing memoria FLASH

- Processo di costruzione del circuito
 - Testing Parametrico di processo
 - I EWS (Electric Wafer Sorting)
 - II EWS
- Assemblaggio
 - Final Test
 - Campionamento per valutazione affidabilistica



Resa di un dispositivo

- La **RESA** di un dispositivo su lotto è il rapporto tra i pezzi funzionanti e i pezzi disponibili
- La resa dipende dalla difettosità del processo
- Per migliorare la resa di un dispositivo ci sono 2 vie:
 - Riduzione della difettosità del processo
 - Introduzione di accorgimenti di design che permettono di aumentare la resa:
 - **Ridondanza**: celle aggiuntive utilizzabili come "scorta", fissate a livello di EWS
 - **Error Correction Codes**: algoritmi in grado di riconoscere e riparare celle di memoria difettose durante la vita del dispositivo



Resa e ridondanza: analisi statistica

- Resa di un dispositivo in assenza di ridondanza:

- p : probabilità che una cella sia difettosa
 - Nr : numero di righe della matrice
 - Nc : numero di colonne della matrice
- $$Y_0 = (1-p)^{Nr \cdot Nc} \Rightarrow p = 1 - Y_0^{1/Nr \cdot Nc}$$

- Resa di un dispositivo con $Nrid$ colonne di ridondanza:

- p : probabilità che una cella sia difettosa
- qc : probabilità che una colonna non contenga bit difettosi
- pc : probabilità che una colonna contenga almeno un bit difettoso
- Nr : numero di righe della matrice
- Nc : numero di colonne della matrice

$$qc = (1-p)^{Nr} \quad Y_{Nrid} = \sum_{m=0}^{Nrid} \left\{ \binom{Nc}{m} \cdot pc^m \cdot qc^{Nc-m} \cdot \sum_{j=m}^{Nrid} \left[\binom{Nrid}{j} \cdot qc^j \cdot pc^{Nrid-j} \right] \right\}$$

$$pc = 1 - qc$$



Error Correction Code

- Gli **Error Correction Codes** sono sistemi basati sulla codificazione dei dati e su algoritmi in grado di riconoscere ed riparare automaticamente celle che cambiano stato logico (es. per problemi di ritenzione)
- Codifica ECC**
 - Quando vengono scritte in matrice le informazioni del cliente, l'algoritmo ECC provvede a scrivere altri bit aggiuntivi (bit di parità) accessibili solo al sistema
- Decodifica ECC**
 - Quando il cliente legge i dati della memoria, l'algoritmo ECC confronta i dati con le informazioni dei bit di parità e nel caso trovi differenze è in grado di fornire una risposta corretta riconoscendo i bit che hanno cambiato stato logico
 - Eventualmente la memoria può anche autocorreggersi, riprogrammando il dato inizialmente memorizzato sui bit difettosi



ECC vs. Ridondanza

ECC

- correggono problemi di affidabilità durante la vita del dispositivo
- occupano area elevata in matrice
- necessitano di circuiteria aggiuntiva per la gestione degli algoritmi
- rallentano il tempo di accesso ai dati

Ridondanza

- occupa poca area in matrice
- risolve alcuni tipi di difettosità delle celle di memoria
- può essere utilizzata a livello di colonna, riga, settore
- viene fissata a livello di EWS, non può essere modificata durante la vita del

ECC e Ridondanza vengono utilizzati insieme per aumentare sia la **resa** che l'**affidabilità**



Memorie Flash multilivello

Partendo dalla relazione che lega la soglia di una cella alla carica nella floating gate

$$\Delta V_T^{cell}(Q_{FG}) = -\frac{Q_{FG}}{C_G}$$

si osserva come il transistor a floating gate non sia intrinsecamente un oggetto digitale, ma sia piuttosto un elettrometro analogico.

Perché non sfruttare quindi queste caratteristiche per memorizzare all'interno di una sola cella di memoria floating gate un maggior numero di informazioni?

SINGLE BIT CELL:
1 cella
2 stati logici
(0 - 1)



MULTI BIT CELL:
1 cella
2ⁿ stati logici
(000 - 001 - 010 - 011
100 - 101 - 110 - 111)

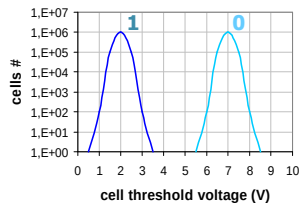
Con una memoria multilivello la densità della memoria aumenta e il costo per bit si riduce (a pari complessità tecnologica!)



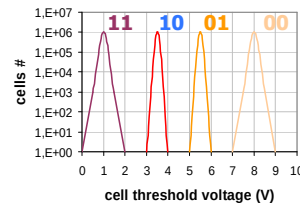
Memoria NOR multilivello (2bit/cell)

Una memoria Flash NOR si presta naturalmente all'introduzione del concetto di multilivello:

1 bit/cell



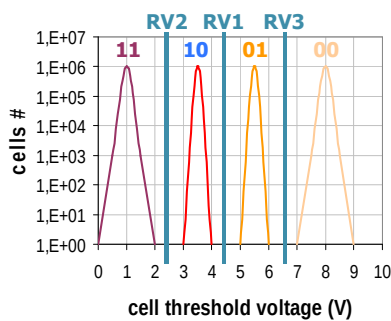
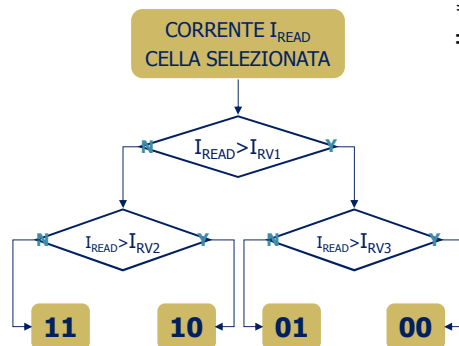
2 bit/cell



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Lettura Flash multilivello (2bit/cell)

- In analogia alla cella single bit, la lettura avviene confrontando la corrente della cella con la corrente **delle 3 celle di riferimento**



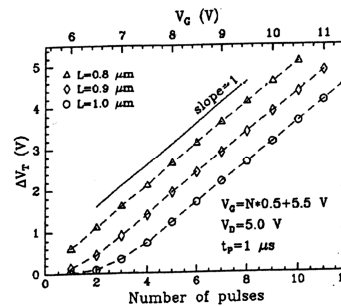
- Dovendo inserire più livelli di sensing, il tempo di accesso risulta in generale aumentato



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Programmazione Flash Multilivello

- La programmazione per C.H.E di una Flash NOR Multilivello è particolarmente delicata poiché bisogna ottenere distribuzioni molto strette e valori di soglia molto controllati per gli stati 10 e 01
- Programmando la cella con una successione di brevi impulsi con tensione di drain costante e tensione di gate crescente, la soglia della cella segue la tensione di gate
- Tra un impulso e il successivo una fase di verifica permette di fermare la programmazione al livello desiderato



Problemi delle Flash multilivello

- Complessità aggiuntiva nei circuiti di sensing
- Minor velocità di lettura e programmazione
- Minori margini sulle distribuzioni
 - Maggior criticità delle dispersioni di processo
 - Maggior sensibilità ai disturbi
 - Minori margini sulla ritenzione
 - Necessità di algoritmi ECC

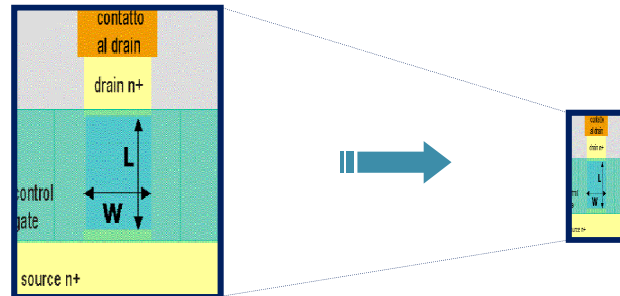
NOR Flash: 180nm → 65nm

λ : technology generation minimum size (i.e.: 180nm)

Scaling rules for NOR Flash cells: area $\approx \sim 10 \cdot \lambda^2$

180nm NOR: 0.326 μm^2

65nm NOR: 0.042 μm^2



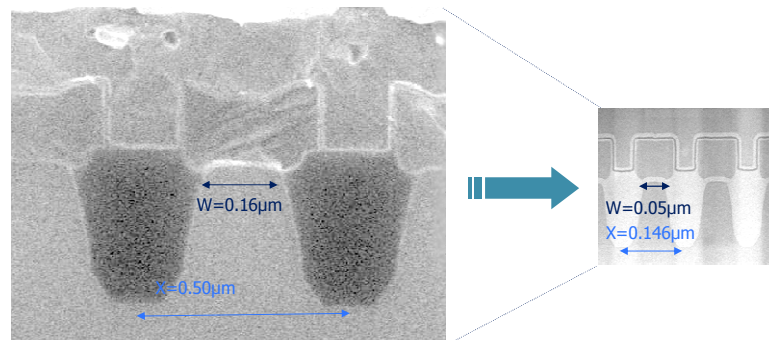
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X-direction: 180nm → 65nm

- Cell scaling along X direction: Shallow Trench Isolation (STI) and width must be reduced

180nm NOR: 0.326 μm^2

65nm NOR: 0.042 μm^2

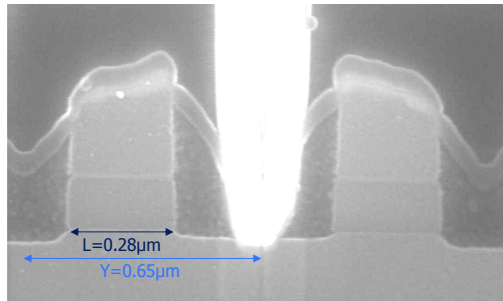


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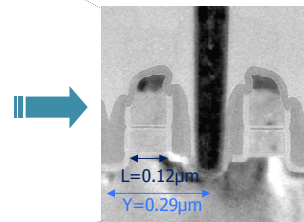
Y-direction: 180nm $\rightarrow$ 65nm

- Scaling along Y direction: cell length (L) must be reduced as well as contact sizes, distances between contacts and WL, and between sources

180nm NOR: 0.326 μ m²

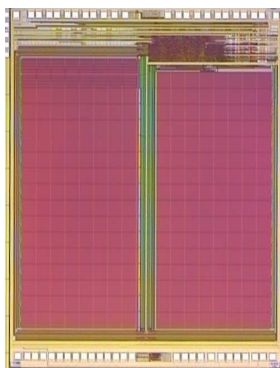


65nm NOR: 0.042 μ m²

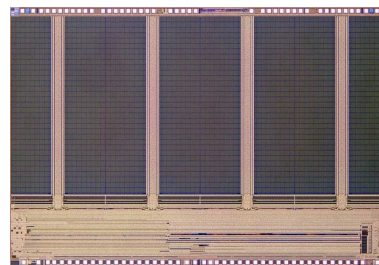


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NOR Flash scaling: an example



**16Mbit Flash 3.0V
Technology 250nm
chip size=28 mm²**

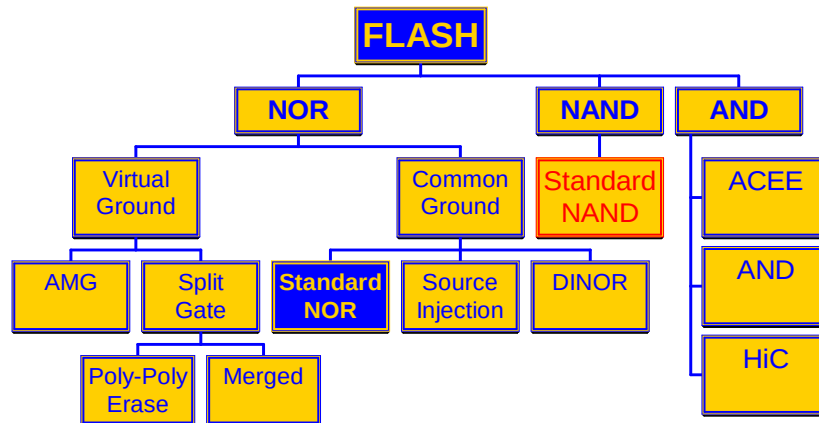


**512Mbit Flash 1.8V
Technology 65nm 2bit/cell
chip size=29 mm²**



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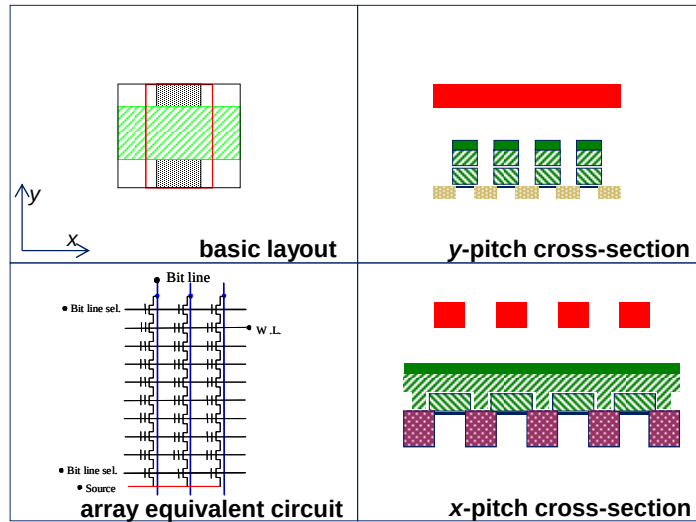
La famiglia delle memorie FLASH



Ideal Mass Storage Technology

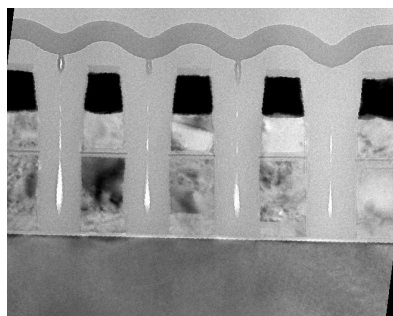
- The **larger capacity** at the **lower cost** (per megabyte) with
 - Strong Ruggedness
 - Low Power Consumption
 - Small Size
 - Light Weight
 - High Reliability
 - Noise immunity
 - Good Performances (High Program and Read Throughput)

NAND Flash

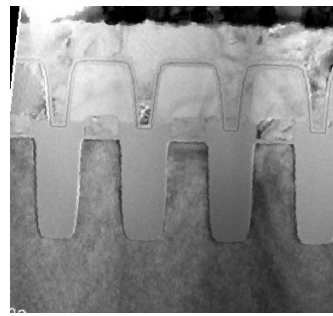


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NAND Cell Cross-section



y-pitch cross-section



x-pitch cross-section

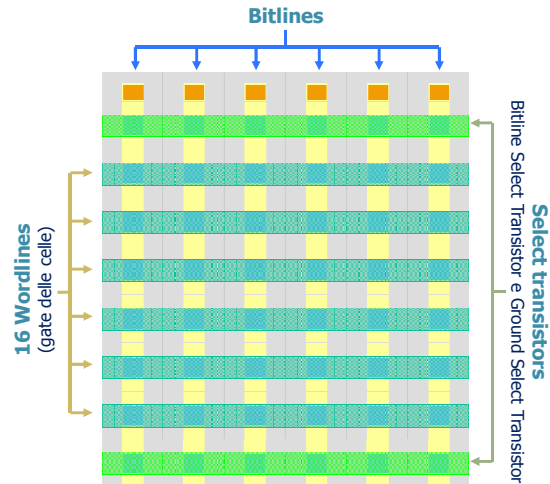
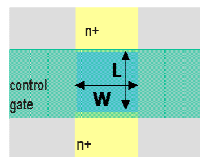
- Cell distance is $2F$ in both directions $\rightarrow 4F^2$ cell size!
- Very simple cell structure $\rightarrow$ easier scaling



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Matrice di memoria Flash NAND

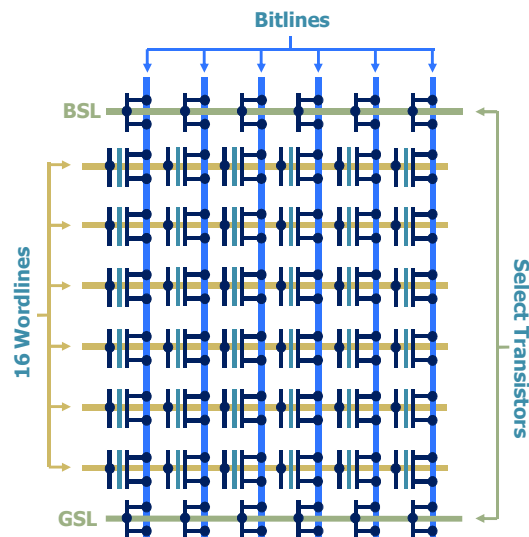
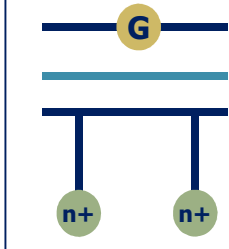
Cella Flash NAND
singola



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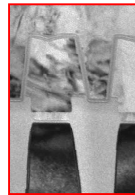
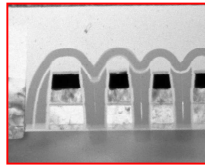
Organizzazione di una Flash NAND

Cella Flash NAND
singola



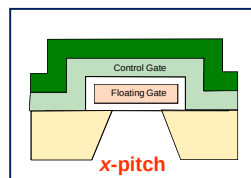
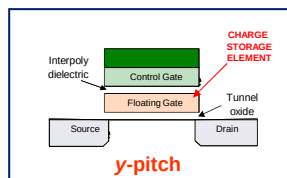
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NAND Stacked Gate Flash



NAND cell

- Tunnel oxide th.: 7-8nm
- ONO EOT: 15nm
- Cell gate length: 130nm
- Cell size: 0.09um²

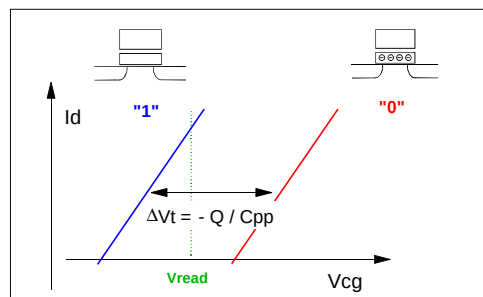


130nm Technology Node



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Reading Operation



"1" $\Rightarrow I_{read} > 0$
 "0" $\Rightarrow I_{read} = 0$

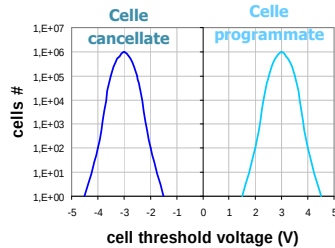
NAND Flash

- Read current: $I = 300-500\text{nA}$
- Random access: $t = 10-30\mu\text{s}$
- Serial throughput: 10-30MB/s



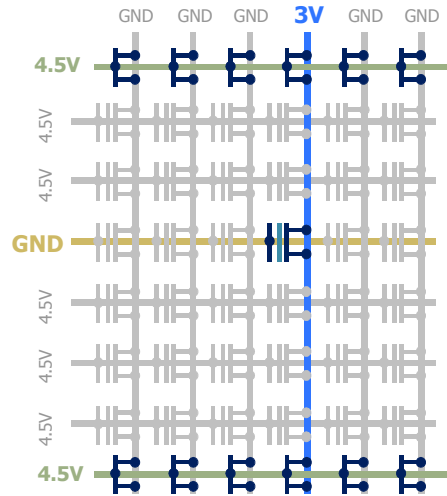
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NAND: lettura



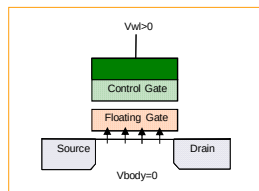
Lettura cella NAND

Dato che tutte le celle non selezionate sono accese, indipendentemente dal loro stato, sulla bitline selezionata passa corrente solo se la cella selezionata è cancellata.



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NAND Flash Writing Mechanism

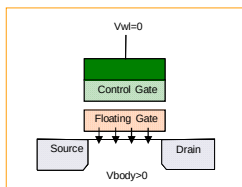


- Programming:**
Fowler-Nordheim (FN) electron tunneling current through the tunnel oxide to the floating gate

V_{wl}	18-20V
V_{body}	0V
t_{pulse}	300us
$I_{current}$	~0
Throughput	7-10MB/s

Threshold voltage range (V): $-5 < V_t < 3$;

Threshold voltage shift (V): $\Delta V_t > 3$



- Erasing:**
Fowler-Nordheim (FN) electron tunneling current through the tunnel oxide from the floating gate to the silicon surface

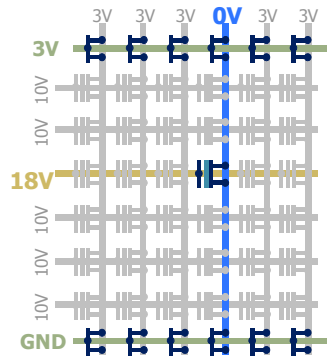
V_{wl}	0V
V_{body}	18-20V
t_{pulse}	2ms
$I_{current}$	~0



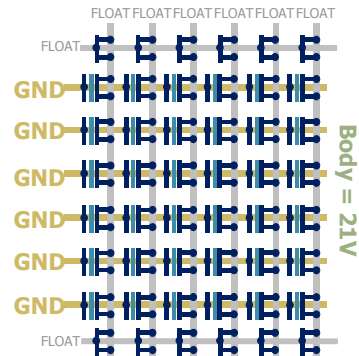
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NAND: Program e Erase

La **programmazione** di una Flash NAND avviene per FN, polarizzando la wordline a tensioni elevate con substrato a massa



La **cancellazione** di una Flash NAND avviene per FN, polarizzando il substrato a tensioni elevate e tenendo tutte le wordlines a massa



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Performances: Data Throughput



**SEQUENTIAL
PROGRAM
7MB/s Max**

Direct Video
Recording



**NAND
FLASH**

**ERASE
64MB/s Max**

1Gbit Chip Erase
in 2 seconds



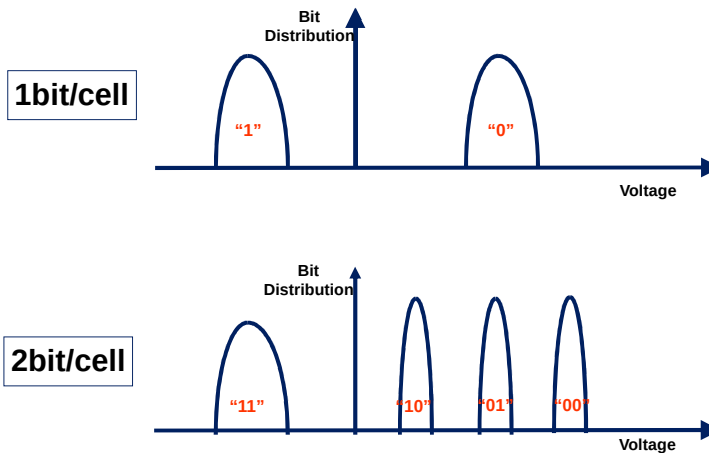
**SEQUENTIAL
READ
27MB/s Max**

Maximum throughput referred to NAND Family X16 with 2Kbytes Page Size w/o Host Overhead



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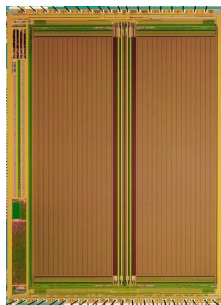
NAND Multi-Level Concept



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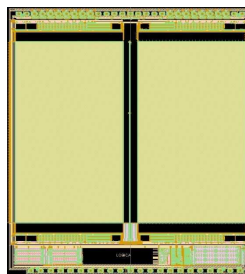
NAND Flash Memory Product

120nm NAND Technology
0.062 μm^2 Cell Size



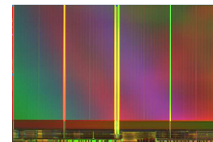
512Mb NAND Flash Memory

90nm NAND Technology
0.038 μm^2 Cell Size



1Gb NAND Flash Memory

25nm NAND Technology
0.0034 μm^2 Cell Size



64Gb 3b/c NAND Flash Memory



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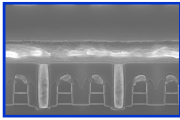
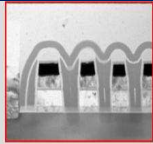
Confronto Flash NAND e Flash NOR

- ↑ La cella NAND è più piccola
 - Non è necessario il contatto tra i drain delle celle
 - La lunghezza di canale è minore poichè non serve tensione elevata al drain
 - Non c'è necessità di realizzare una giunzione graduale al source
- ↑ La memoria NAND consuma meno corrente
 - I meccanismi di scrittura per FN sono più efficienti del C.H.E.
- ↓ Il tempo di accesso random di una NAND è molto più lento
 - La lettura "in serie" riduce molto la corrente disponibile
- ↓ La memoria NAND necessita di tensioni più elevate
 - I meccanismi di scrittura FN richiedono forti campi elettrici
- ↓ La memoria NAND è più sensibile ai disturbi di programmazione



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NOR and NAND Stacked Gate Flash

	 NOR	 NAND
Cell size (F²)	10	5
Read access	Random (fast ~50ns)	Serial
Progr. mech./ throughput	CHE / 0.5 MB/s	FN / 8-10MB/s
SEM Cross-section (BL direction)		



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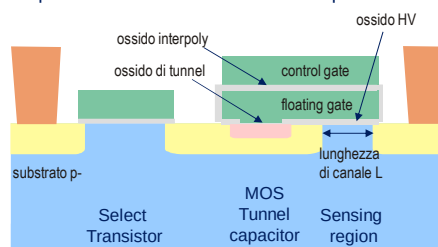
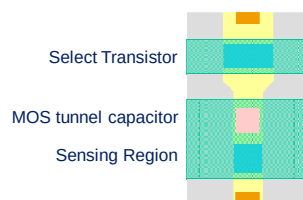
NOR-NAND Architecture Comparison

- Common Cell Architecture:
 - Floating Gate Concept
 - One-Transistor Stacked-Gate Cell
- Different Transistor Architecture:
 - High Performance Logic in NOR:
 - To speed the program/erase algorithm
 - To get the fastest random access time
 - Dedicated Logic in NAND, driven by the Cell Architecture
 - To minimize the mask number
 - To reduce the process cost
- Different Memory Reliability Requirement
 - NOR, after Final Test, must be a perfect array (100% functionality)
 - NAND is similar to a mass storage media (fault tolerant, like HD):
 - ECC (64bit every 512)
 - 98% array functionality (2% of bad blocks on field admitted)

La cella EEPROM

EEPROM Electrically Erasable and Programmable ROM

- Sia la programmazione che la cancellazione avvengono per FN tunneling sulla regione del condensatore di tunnel
- La lettura della cella avviene sul transistor di sensing
- **La memoria EEPROM è programmabile e cancellabile a livello di byte, grazie ai transistori di selezione**
- **Rispetto alla cella EPROM o FLASH, la cella EEPROM è molto più grande** a causa del select transistor e della separazione tra la zona di scrittura e quella di lettura



Conclusions

- In the last decade the NVM market increased exponentially due to the requirements of mobile applications and portable systems
- Floating gate concepts has been proven to be a very reliable mechanism for Flash memory fabrication
- Flash memories are expected to be the mainstream NVM for the next years
- NOR Flash is the preferred option for code storage due to their high performances
- NAND Flash is the preferred option for data storage due to their very low cost



December 11

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Corso di Laurea in Ingegneria Elettronica - Università degli Studi di Udine

Tecnologia e Progettazione di **MEMORIE NON VOLATILI**

Agostino Pirovano

Roberto Bez

Alessandro Grossi

Giorgio Servalli

Process R&D

Micron

Agrate Brianza (Milan), Italy



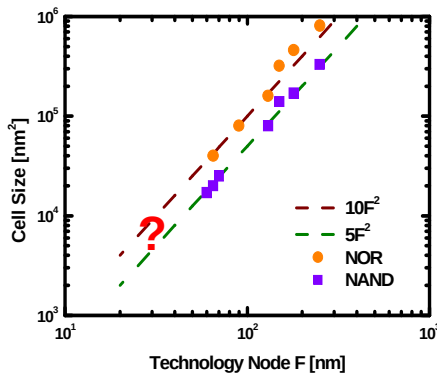
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- PROBLEMI DI SCALABILITA' DELLE MEMORIE FLASH
 - Elementi attivi
 - Elementi passivi
- ALTRE MEMORIE NON VOLATILI
 - FERAM
 - MRAM
 - RRAM

- AN OUTLOOK INTO THE FUTURE
 - Una lezione sullo scaling
 - Memorie a cambiamento di fase (PCM)
 - Possibili evoluzioni delle memorie PCM
 - Memorie a cross-point

Flash Cell Evolution



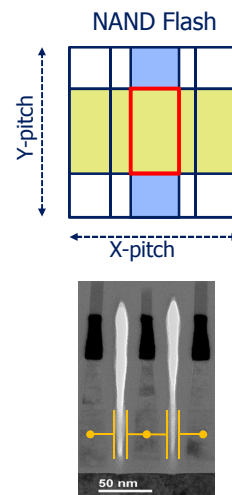
- Flash cell size reduction following the Moore's law
- Cell basic structure unchanged through the different generations
- Scaling beyond the 45 nm technology node for NOR Flash and beyond 22nm for NAND Flash is still considered critical



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Flash Cell Scaling Challenges

- Cell basic structure unchanged through the different generations
- Cell area scaling through:
 - Active device scaling (W/L)
 - Passive elements scaling
- Main scaling issues:
 - Number of stored electrons
 - Cell proximity interference
 - Tunnel and interpoly dielectric thickness
 - Isolation spacing and WL voltage increase
 - Random Telegraph Noise
 - Trapping/detrapping, SILC
 - Retention after cycling



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Innovations in Flash Technology

- System management techniques
 - Charge placement algorithms
 - Error management techniques
 - Multi-level memories
- High-k dielectrics and "discrete-trap" memories
 - Reduced oxide thickness
 - Lower energy barrier height
 - Improved reliability
- Fin-FET and 3D architectures
 - Moves the scaling constraints along the vertical dimension
 - Higher performance

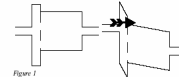
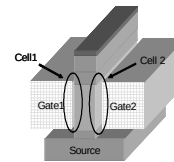
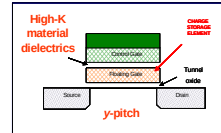


Figure 1. Band diagram of a $\text{SiO}_2/\text{AlO}_x$ stack when no bias is applied (left-hand side) showing a large tunneling barrier, and when a moderate voltage is applied (right-hand side) showing a strongly reduced barrier width. The high dielectric constant of the second layer limits the voltage drop over the barrier, so that a high field in the SiO_2 layer can be obtained for a moderate applied bias.

P. Blomme et al., "A novel low voltage memory device with an engineered $\text{SiO}_2/\text{high-k}$ tunneling barrier", NVSMV 2003



A. Fazio, MRS Bulletin, Nov. 2004

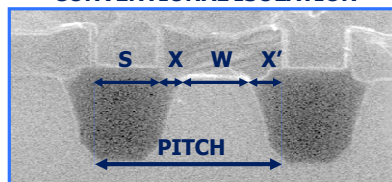


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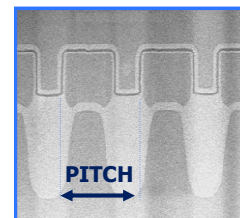
Technology Challenges for Scaling

- Continuous technology innovations are required for Flash memories scaling
 - Advanced lithography for high resolution
 - Light wavelength reduction
 - Methods to reduce diffraction effects must be introduced
 - Thermal treatments reduction
 - Self-aligned process schemes

CONVENTIONAL ISOLATION



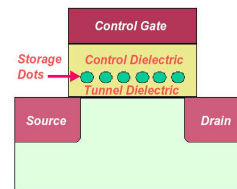
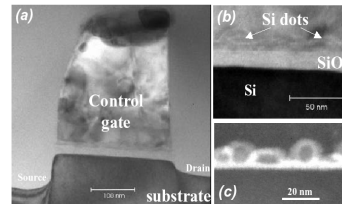
SELF-ALIGNED ISOLATION



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Flash Evolution: Nanocrystal and Charge-Trap (CT) Memories

- Storing mechanism
 - Electrons trapped into silicon nanocrystals or trapping centers that act as nano-floating gates
- Writing mechanism
 - FN or direct tunneling
 - Channel hot electrons
- Sensing mechanism
 - Change in the threshold voltage of a MOSFET
- Cell structure
 - 1 Transistor (Flash-like) structure



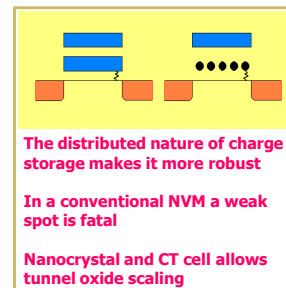
European Project ADAMANT



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Nanocrystal and CT: Advantages and Issues

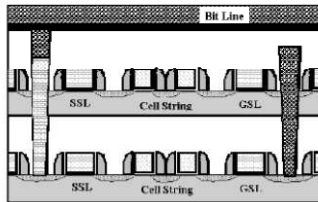
- Main advantages
 - Evolutionary with respect to FG memories
 - Integration and full compatibility with conventional CMOS processes
 - Robustness to parasitic FG cross-talk (interference coupling)
 - Robustness to stress induced leakage current (SILC)
- Main issues
 - Low threshold voltage shift (<3 V)
 - Retention and endurance characteristics to be deeper investigated
 - Difficult retention-programming trade-off for CT memories
 - Scaling concerns related to nanocrystal distributions



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Samsung stacked NAND-concept

Samsung presented at IEDM 2006 a stacked NAND based on multi silicon layers grown by epitaxy



The integration scheme is based on mono crystal silicon epitaxy

32 bit TANOS-NAND cell string with 63 nm dimension



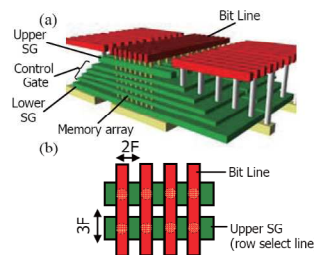
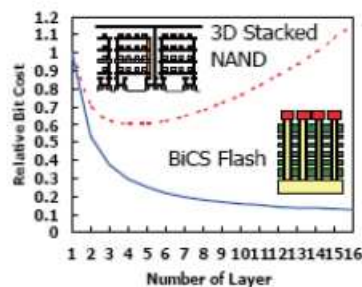
Jung et al., IEDM 2006, pg. 37- 39



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Toshiba 3D approach-concept

Toshiba presented at VLSI 2007 an interesting 90nm 3D approach for Multi-layer TANOS-NAND technology



Number of layers independent from critical steps

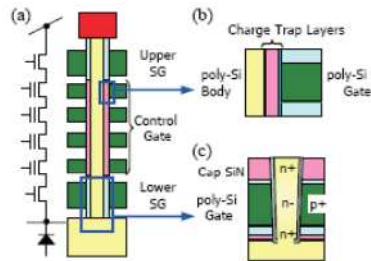
Tanaka et al., VLSI 2007, pg. 14-15



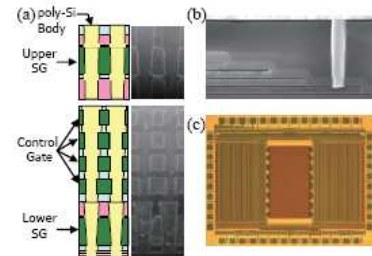
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Toshiba 3D approach-concept

The proposed architecture is very challenging but the process is really inexpensive



Selector transistor and memory stack must be integrated separately (3 critical mask for the NAND STRING and 1 critical mask for the routing). Due to the Overlay constraints the cell size is $6F^2$



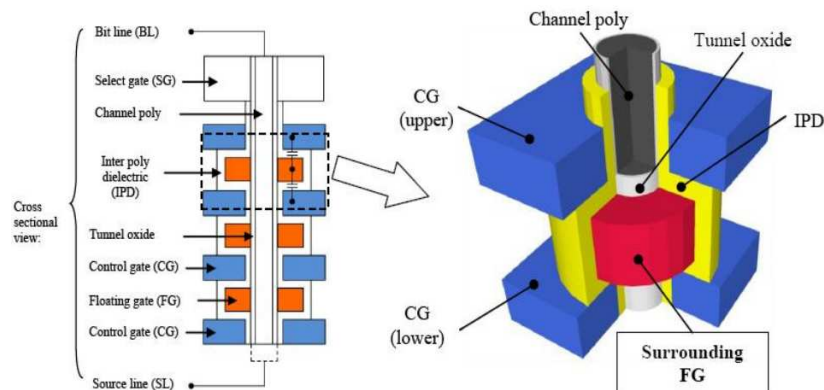
For stacked NAND (3 critical layers) the Cost per bit increases if more than 3 layers are stacked



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Hynix 3D floating-gate Flash

- A 3D vertical NAND Flash with floating gate



S. Whang et al., IEDM (2010)



December 11

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3D NAND Status and Development

- 3D NAND Flash have several concerns
 - All approaches have big process/fabrication issues
 - More complicated P/E procedures (hole injection, ...)
 - Disturbs are increased due to shared electrodes
- It is a well-defined problem (in particular the FG approach) that can be effectively addressed by semiconductor industries



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Flash Limitations

- Limited endurance (10^5 cycles)
- Slow operations
 - NOR slow write ($\sim 5\text{-}10\mu\text{s}/\text{byte}$ program, $\sim 1\text{ sec}/\text{Mbit}$ erase)
 - NAND slow random read ($30\mu\text{s}$)
- Data flexibility
 - NAND page program
 - NAND and NOR sector erase
- Cell scalability beyond 40 nm (in particular for NOR Flash)
- Difficult process architecture with high-voltage devices for program and erase operation



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Key Requirements of an Alternative NVM

- Readiness for beyond leading edge technology node
- Scalability
- Cost structure
 - MLC capable
 - 3D stackable
- Performance
 - High Program and Read Throughput
 - Low power
 - Flexibility
- Reliability
 - Non-volatility with long retention (e.g. > 10 years)
 - Extended number of read cycles
 - High program endurance



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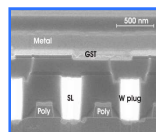
Near-Term and Long-Term Alternatives

More than 35 NVM alternatives have been so far proposed...

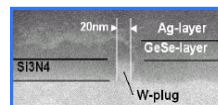
FERAM



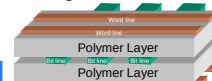
PCM



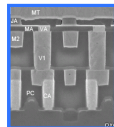
PMC RRAM



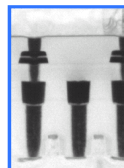
Polymer FeRAM



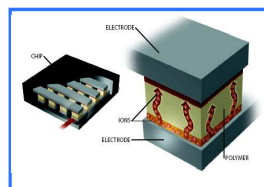
MRAM



MOx-RRAM



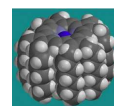
Polymer RRAM



CNT



Molecular



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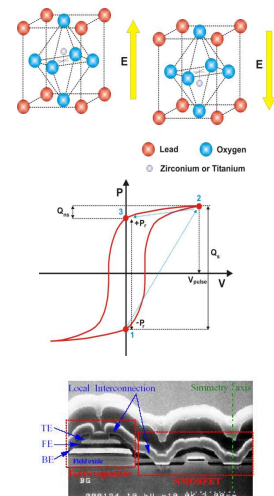
NVM Categories

- **Electronic decoded, lithography dependent (Moore's law follower)**
 - Transistor selected (like DRAM or Flash)
 - Ferroelectric memory (FERAM)
 - Magnetoresistive memory (MRAM and STT-MRAM)
 - Resistive RAM (RRAM)
 - Phase-Change Memory (PCM)
 - Cross-point memories (Passive arrays)
 - Ferroelectric polymers (PFRAM or TFEM)
 - Organic charge-transfer complexes (conductive polymers)
 - Resistive switching
- **Mechanical decoded, lithography independent (beyond Moore's law)**
 - Probe storage (Seek and Scan, like Hard Disk or CD)
 - Polymers
 - Chalcogenide
 - Ferroelectric



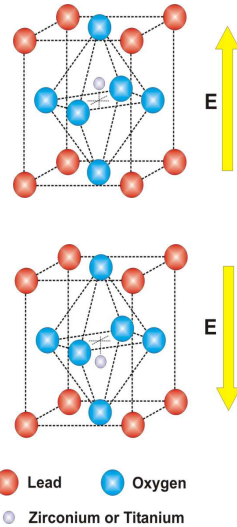
Ferroelectric RAM (FeRAM)

- **Storing mechanism**
 - Permanent polarization of a ferroelectric material
- **Writing mechanism**
 - Electric field produced in the ferroelectric layer by the voltage applied to the capacitor plates
- **Sensing mechanism**
 - Displacement current associated to the polarization switch
- **Cell structure**
 - DRAM-like: 1 transistor, 1 capacitor (1T/1C)



Ferroelectric Materials

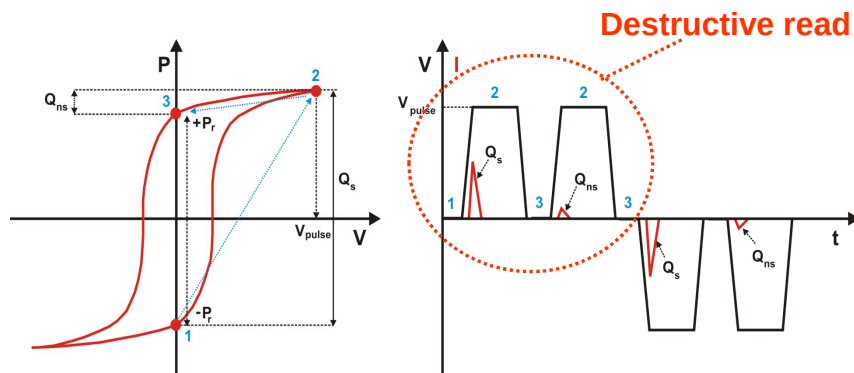
- **Ferroelectric materials exhibit, over some range of temperature, a spontaneous electric polarization that can be oriented by application of an electric field**
- **Main ferroelectric thin film materials**
 - ▶ PZT: Lead-Zirconate-Titanate $\text{PbZrTi}_{1-x}\text{O}_3$
 - ▶ SBT: Strontium-Bismuth-Tantalate $\text{Sr}_{1-y}\text{Bi}_{2+x}\text{Ta}_2\text{O}_9$
 - ▶ BLT: La substituted-Bismuth-Titanate $\text{Bi}_{4-x}\text{La}_x\text{Ti}_3\text{O}_{12}$
- **Electrical characteristics**
 - ▶ High P_r value (10-30 $\mu\text{C}/\text{cm}^2$)
 - ▶ Switching voltage ($\sim 1.5\text{-}3\text{ V}$)
 - ▶ Reliability
- **Technological characteristics**
 - ▶ Formation temperature (usually higher than 600° C)
 - ▶ Electrode interaction (P_r and integration issues)
- **MOCVD PZT dominates for scaled technologies**



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Operating Principles

The basic memory element is a ferroelectric capacitor (FeCAP)

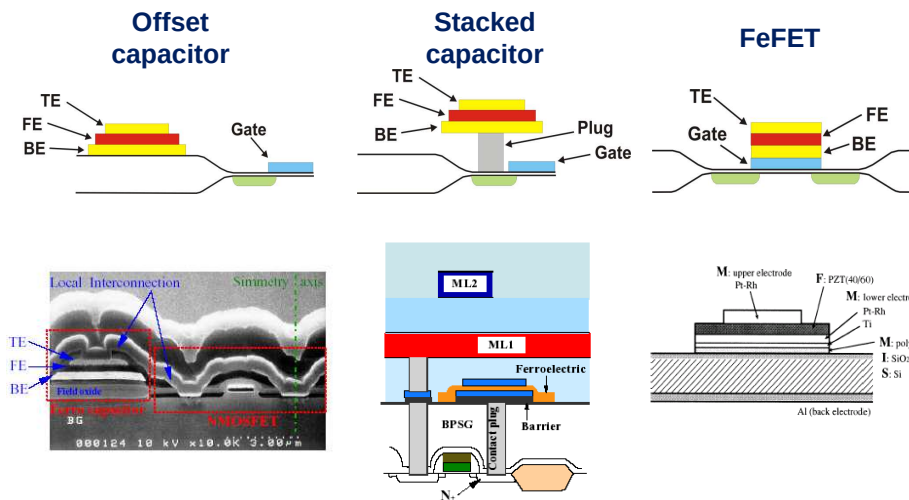


Read out signal: $Q_s - Q_{ns} = 2P_r$



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FeCAP Architectures



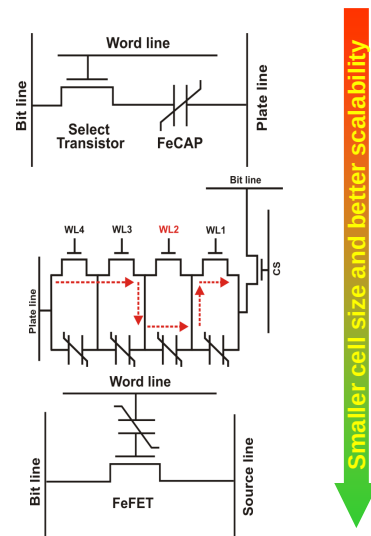
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FeRAM Cell Architectures

2T2C and 1T1C cell
(DRAM architecture)

Chain-type FeRAM
(NAND Flash architecture)

FeFET
(MOFET-like Flash architecture)

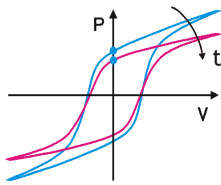


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FeRAM Reliability

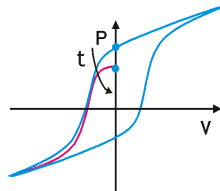
**Endurance
(electrical
fatigue)**

**Polarization
loss
upon cycling**



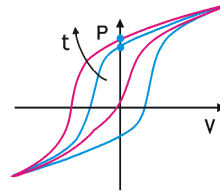
Retention loss

**Time dependent
polarization loss**



Imprint

**Shift of the
hysteresis loop**



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FeRAM Scaling

Planar FeCAP area scaling

Minimum capacitance for sensing: 30 fF
Operating voltage: 1 V

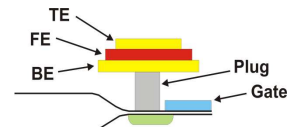
Minimum charge for sensing: 30 fC
Equivalent electron number: 200000

Technology node: 90 nm
FeCAP area: $100 \times 100 \text{ nm}^2$
 P_r : $20 \mu\text{C}/\text{cm}^2$

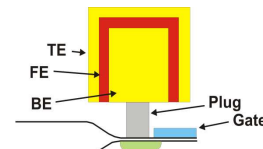
Available electron number: 25000



3D approach necessary!



2D FeCAP



3D FeCAP



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FeRAM: Advantages and Issues

- Main advantages
 - Fast (<100ns) read and write operations with no intrinsic limitation (<100ps)
 - High write endurance (>10¹²)
 - Low voltage and low power operation
- Main issues
 - Limited read endurance, destructive read-out (apart FeFET)
 - Difficult process integration
 - Large cell size vs. Flash and DRAM (>15F²)
 - Scaling limits and 3D capacitor required to go beyond the 90nm technological node



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FeRAM Development Status

Company	Fujitsu	Matsushita	Samsung			TI/Ramtron	Toshiba
Source	IEDM'02	VLSI'04	VLSI'05	IEDM'05	IEDM'06	VLSI'03	ISSCC'06
Technology Node [nm]	180	180	150	180	150	130	130
Density	4Mb	1Mb	64Kb	2Mb	64Mb	64Mb	64Mb
Cell size [μm ² -F ²]	1.3-40	2.4-74	0.27-12	0.48-15	0.34-15	0.54-32	0.61-36
FeCAP size [μm ² -%]	0.49-38%	-	0.11-40%	0.26-54%	0.16-47%	0.25-46%	0.2-32%
FE Material	MOCVD PZT	SBT	MOCVD PZT	MOCVD PZT	MOCVD PZT	PZT	PZT
2P _r [μC/cm ²]	31	-	35	38	40	24	42
Operation voltage [V]	1.8	2.7	<1.2	1.6	1.8	1.3	1.8
Cycle time [ns]	-	2000	-	60	100	35	60

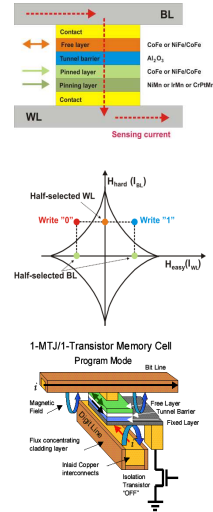
Low-density stand-alone (up to 4Mb) and embedded products on the market for a number of years in relaxed technologies (0.35 μm) from Fujitsu and Ramtron



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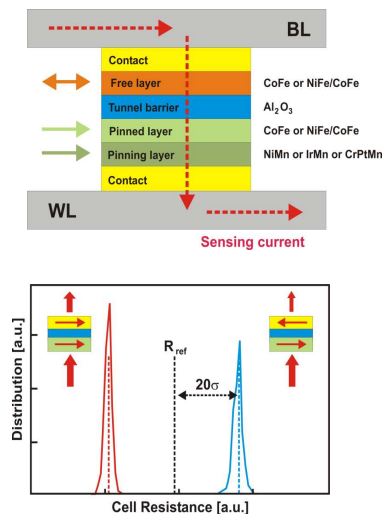
Magnetoresistive RAM (MRAM)

- **Storing mechanism**
 - ▶ Permanent magnetization of a ferromagnetic material in a Magnetic Tunnel Junction (MTJ)
- **Writing mechanism**
 - ▶ Magnetic field produced by the current flowing in the array bit and digit lines
- **Sensing mechanism**
 - ▶ Resistance change in the MTJ
- **Cell structure**
 - ▶ 1 transistor, 1 resistor (1T/1R)



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MTJ Principles

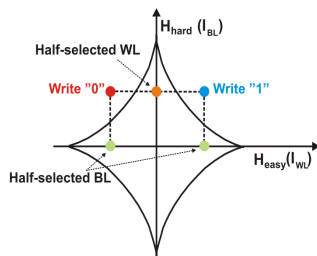
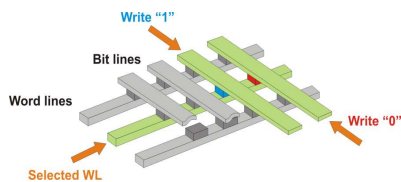


- Magnetic Tunnel Junction constituted by a pinned magnetic layer, an insulator, and a free magnetic layer
- Information stored in the magnetization direction (parallel or anti-parallel) of the free layer
- Read out performed comparing to a reference resistance



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MRAM Writing Principles



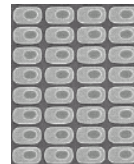
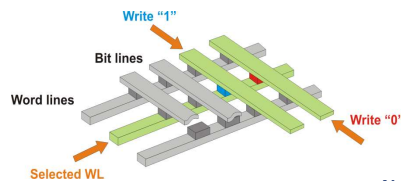
- Writing mechanism
 - Vector sum of magnetic field generated by Digit Line and Bit Line current switch the MTJ free layer (Stoner-Wohlfarth switching)
- Writing disturb
 - Selectivity is based on Astroid diagram: half selected bits must not switch
- Writing current
 - Digit Line and Bit Line current each $\sim 5\text{-}10\text{mA}$
- Key issue
 - Complex bit shape and uniformity



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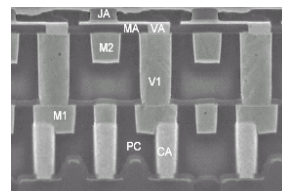
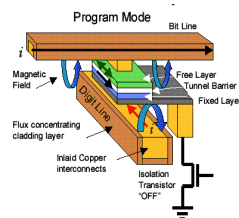
MRAM Cell Architectures

Cross-point architecture



N. Sakimura *et al.*, ISSCC 2003

MOSFET-selected architecture



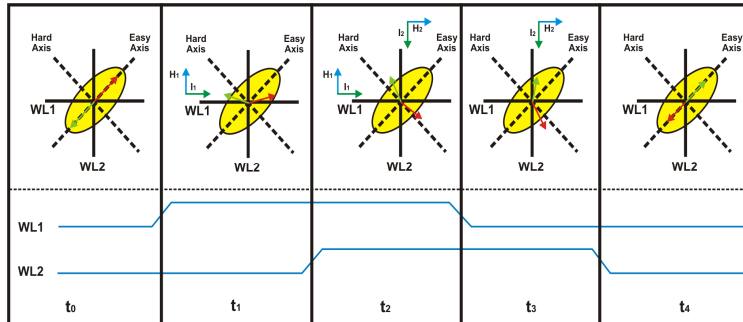
W. J. Gallagher, Taiwan NVM Workshop, 2005



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Toggle MRAM

Savtchenko switching "Toggle": antiparallel bilayer free layer oriented 45° with respect to the write wires, written by rotating the magnetic field



M. Durlam et al., IEDM Tech. Dig., 2003

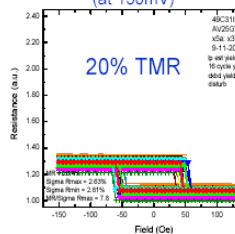
Program disturb issue is resolved, but a larger programming current is required (~7mA vs. ~4mA of SW MRAM)



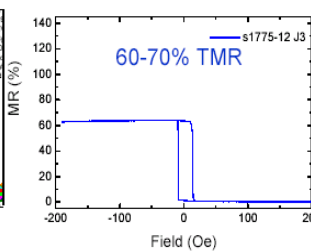
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MgO-based MTJ

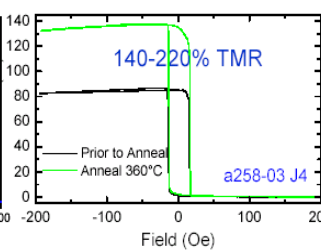
1995
Al₂O₃ tunnel barriers
NiFe alloy electrodes
(at 150mV)



~2001
Al₂O₃ tunnel barriers
CoFe(B) electrodes



2001-2002
MgO tunnel barriers
CoFe(B) electrodes
(internal results)



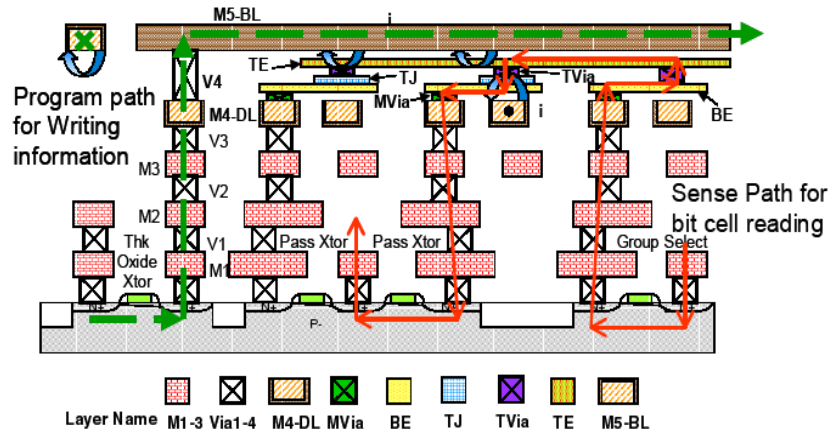
A larger Tunneling Magnetoresistance Ratio (TMR) can be achieved in MgO-based MTJ devices (up to 220% at room temperature)

W. J. Gallagher, Taiwan NV Memory Workshop, 2005



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Interconnections Scheme



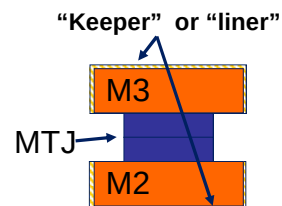
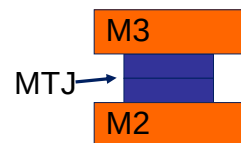
M. Durlan et al., IEDM Tech. Dig. 2003



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MRAM Reliability

- Electromigration of programming metal lines ($\sim 10 \text{ MA/cm}^2$)
- Tunnel barrier dielectric reliability
- Back-end-of-line dielectric deposition and thermal budget
- MTJ stack delamination
- Front-end-of-line contamination due to new material integration
- Soft error rate for thermally activated random switching



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MRAM Scaling

Failure probability

$$P(t) = 1 - e^{-\frac{t}{\tau} e^{-\frac{E_b + E_{b_k}}{k_B T}}} \leq 1 \text{ppm@}85^\circ\text{C for 10 years}$$

Thermal stability

$$E_{b_1} \propto kV \quad \rightarrow \quad \text{Higher k or larger volume V}$$

$$E_{b_2} \propto AR - 1 \quad \rightarrow \quad \text{Higher aspect ratio AR}$$

Power consumption

$$H_{\text{write}_1} \propto k \quad \rightarrow \quad \text{High programming current}$$

$$H_{\text{write}_2} \propto AR - 1$$



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MRAM Scaling

- Thermal stability
 - Smaller cell volume implies a lower thermal stability
 - Compromise between power consumption and thermal stability
- Power consumption
 - High power consumption as large magnetic fields required for switching (~4mA in standard MRAM and ~7mA for Toggle MRAM)
 - Power consumption will increase upon scaling due to cell shape anisotropy

**1st generation not expected to be viable
beyond the 90 nm technology node**



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MRAM: Advantages and Issues

- Main advantages
 - Fast write (<100 ns)
 - High write endurance
 - Low voltage write
- Main issues
 - Difficult process integration
 - Large cell size vs. Flash and DRAM
 - Large write current (> 10 mA/B)
 - Small read signal
 - Scaling limits



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MRAM Development Status

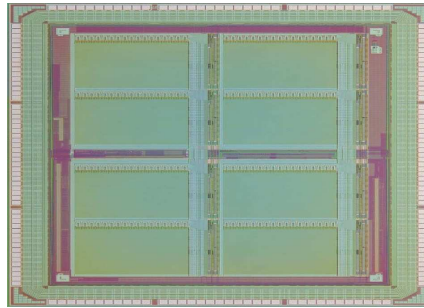
Company	Motorola	Renesas	Nec Toshiba	TSMC	Samsung	IBM Infineon	Freescale STM Philips	Hitachi
Source	ISSCC'04	VLSI'04	IEDM'04	IEDM'04	VLSI-TSA'05	VLSI-TSA'05	VLSI-TSA'05	ISSCC'07
Technology Node [nm]	180	130	130	180	240	180	90	200
Density	4Mb	1Mb	1Mb	1Kb	64Kb	16Mb	4Kb	2Mb
Cell size [μm^2 -F ²]	1.55-48	0.81-48	0.1152-7	1.49-46	-	1.42-44	0.29-36	2.56-64
Cell type	1T-1MTJ	1T-1MTJ	1MTJ	1T-2MTJ	1T-1MTJ	1T-1MTJ	1T-1MTJ	SP-MTJ
Current/bit [mA]	9	-	4	2.5	-	5	-	0.2
Operation voltage [V]	1.8/3.3	1.2	1.5	1.8	-	1.8/2.5	-	1.8
Cycle time [ns]	-	-	250	40	-	30	-	100

Freescale Semiconductor Inc. started commercial shipment of 4Mb MRAM (39F² cell size, 35 ns cycle time) in July 2006



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MRAMs today



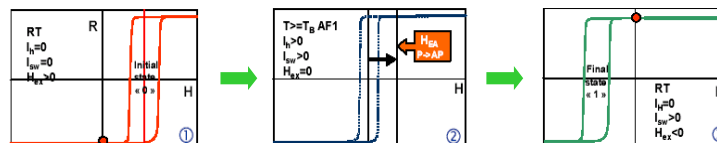
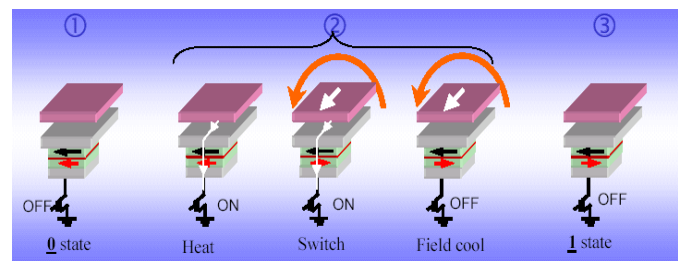
4 Mb MRAM die,
Freescale, now Everspin
(16 Mb available)

- 0.18 μm CMOS with 3 layers of Al and 2 layers of Cu interconnects
- Cladded write lines
- 3.3 V supply voltage
- Symmetrical 35 ns read and write timing
- Cell size = $1.55 \mu\text{m}^2$ (48 F²)
- Die size 4.5 x 6.3 mm²



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Thermal-Assisted MRAM (TA-MRAM)



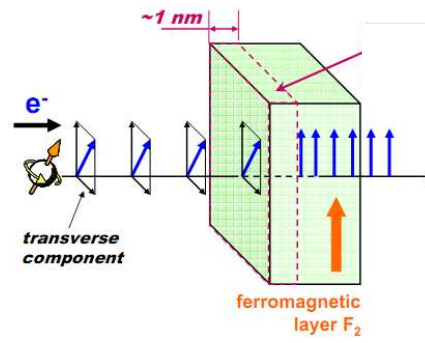
R. Sousa *et al.*, EPCOS 2006



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Spin Transfer Effect

- Incoming electrons lose transverse component of spin in the ferromagnetic layer
- Momentum conservation implies a torque applied by electron current
- Over a threshold J_c , current can switch magnetization



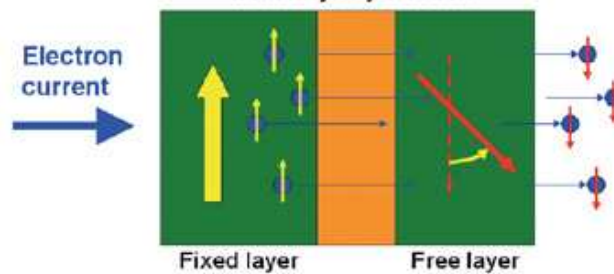
C. Chappert lecture (2008)



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Spin transfer torque: write "0"

a) Writing a bit "0": Spin momentum transfer from majority electrons



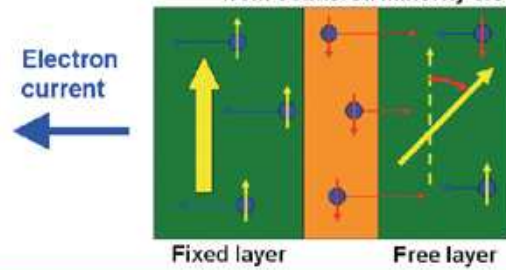
- Two ferromagnetic layers separated by a non-magnetic metallic spacer
- Fixed layer spin-polarizes the current
- The torque switches the free layer magnetization



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Spin transfer torque: write "1"

b) Writing a bit "1": Spin momentum transfer from scattered minority electrons

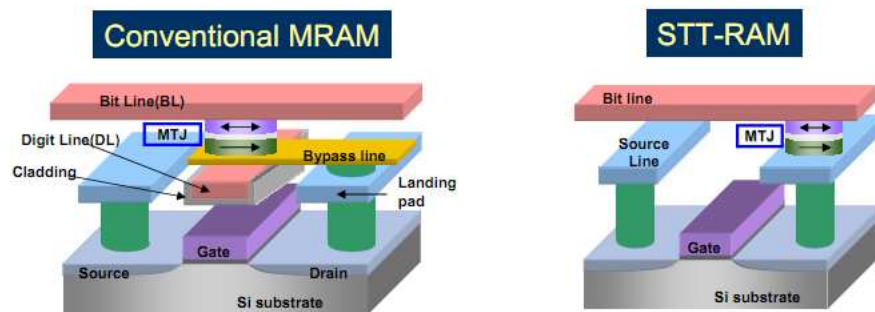


- Electrons injected from the free layer
- Spin-dependent scattering reflects electrons with spin opposite to fixed layer magnetization
- Torque exerted by reflected electrons



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STT MRAM cell



- No digit line, no cladding $\Rightarrow$ ideally, a $6F^2$ cell
- Potentially scalable to 20 nm

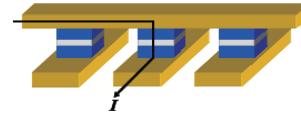
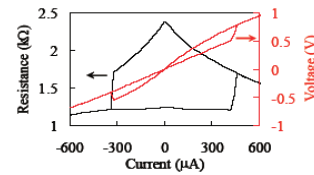


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SPRAM or STT-MRAM (Spin-transfer torque RAM)

- **Basic concept**

- ▶ In a conventional MRAM, the parallel or anti-parallel configuration is formed by applying a cross-point synthetic field induced by a current passing through bit/word lines
- ▶ SPRAM uses the current-induced switching caused by spin-transfer torque

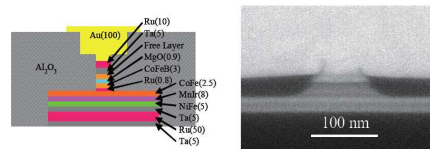


- **Advantages**

- ▶ Cell size: $4F^2$, but limited by CMOS selector to $4F^2$
- ▶ Can mitigate some MRAM issues

- **Issues**

- ▶ Self-read disturbance
- ▶ Writing time depends on the device area
- ▶ Integration



K. Miura *et al.*, VLSI Symp. on Tech. 2007



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Low-power and Embedded Applications

- **FeRAM advantages**

- ▶ Fast (<100 ns) and low energy write
- ▶ Low voltage and low power operation

- **FeRAM Main issues**

- ▶ Large cell size vs. Flash and DRAM
- ▶ Scaling limits

- **Small densities low voltage and low power high performance (niche) market**

- ▶ Contactless smartcards and ID tags
- ▶ Ultra low-power applications

- **MRAM advantages**

- ▶ Fast (<100 ns) read and write
- ▶ Very high write endurance
- ▶ Low/mid voltage operation

- **MRAM Main issues**

- ▶ Large cell size vs. Flash and DRAM
- ▶ Scaling limits (better situation for STTMRAM)

- **Small densities embedded high performance market**

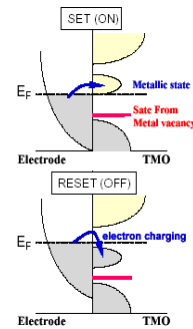
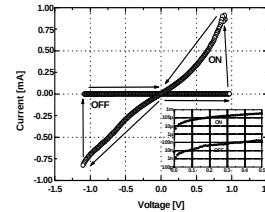
- ▶ Micro and embedded applications
- ▶ Portable and battery operated high performance systems



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Resistive RAM (RRAM)

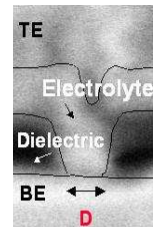
- **Storing mechanism**
 - Resistive switching of a storage layer
- **Writing mechanism**
 - Current or voltage-induced conductance switching
- **Sensing mechanism**
 - Resistance change
- **Cell structure**
 - 1 transistor, 1 resistor (1T/1R) or cross-point (1R)



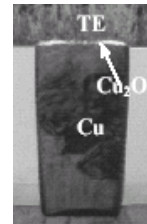
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RRAM Proposed Alternatives

- **Chalcogenide**
 - GST and other phase-change alloys
 - AgGeSe, AgGeS, WO_3 and SiO_2 solid electrolyte
- **Binary oxide**
 - Nb_2O_5 , Al_2O_3 , Ta_2O_5 , TiO_2 , ZrO_x , Cu_xO and NiO
- **Oxides with perovskite structure**
 - $SrZrO_3$, doped- $SrTiO_3$, $Pb(Zr_xTi_{1-x})O_3$ and $Pr_{0.7}Ca_{0.3}MnO_3$
- **Conductive polymers**
 - Bengala Rose, AlQ_3Ag , $Cu-TCNQ$



M. Kozicki, EPCOS 2006



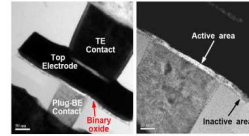
A. Chen *et al.*, IEDM Tech. Dig. 2005



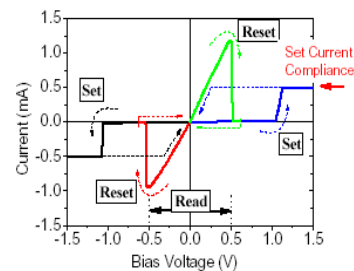
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Resistive Oxides Memories

- **Basic concept**
 - Resistive switching in a binary oxide layer
 - Can be coupled with an oxide based selecting diode
- **Advantages**
 - Cell size: 4-8 F²/n
 - Very low-cost solution
 - Reasonably good endurance and performance
- **Issues**
 - Leakage current
 - Programming current quite high
 - Temperature stability
 - Switching mechanism still controversial



I. G. Baek *et al.*, IEDM Tech. Dig. 2005



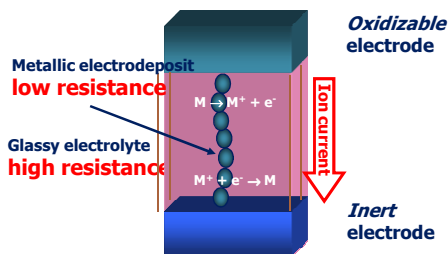
I. G. Baek *et al.*, IEDM Tech. Dig. 2004



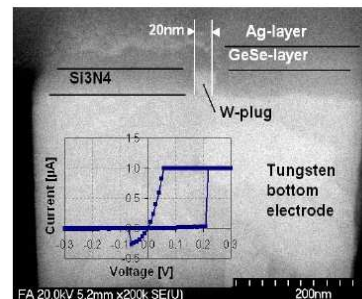
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Programmable Metallization Cells

Programmable metallization cell (PMC): a conductive filament of silver is created by diffusion into a chalcogenide (solid electrolyte) by applying an electric field



M. Kozicki, EPCOS 2006



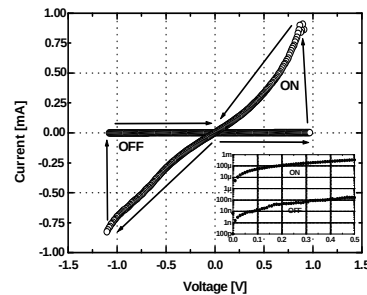
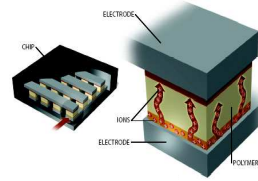
M. Kund *et al.*, IEDM Tech. Dig., 2005



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Polymeric Memories

- **Basic concept**
 - ▶ Storage material located at the cross points
 - ▶ Several polymers proposed for the storage element (Rose Bengal, Fluoresceine-based polymer, DDQ, TAPA, Cu:TCNQ)
 - ▶ Can be coupled with a polymeric selecting diode
- **Advantages**
 - ▶ Cell size: 4-8 F2/n
 - ▶ Very low-cost solution
- **Issues**
 - ▶ Leakage current
 - ▶ Poor performance
 - ▶ Integration and temperature stability



A. Pirovano *et al*, Solid-State Electronics, 2005



RRAM: Advantages and Issues

- | | |
|--|---|
| <ul style="list-style-type: none"> • Main advantages <ul style="list-style-type: none"> ▶ Good read signal window (factor ten in resistance) ▶ Medium/low voltage write ▶ Low programming current and energy ▶ Cross-point solutions available ▶ Good scalability | <ul style="list-style-type: none"> • Main issues <ul style="list-style-type: none"> ▶ Low maturity, no multi-Mb test-chip has been so far presented ▶ Difficult process integration (low thermal budget required) ▶ Retention capabilities for 10years at 85° C must be demonstrated |
|--|---|



Key Learnings

- Opportunities exist in the NVM market for new memory concepts that can provide a competitive advantage, but...



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Floating-Gate NVM (Successful) History

1967 → First Floating Gate Structure

1971 → FAMOS

1977 → EPROM

1980 → EEPROM

1985 → 1T EEPROM (Flash)

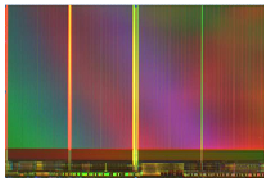
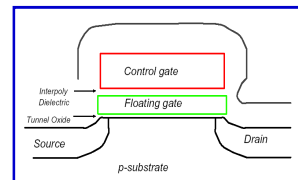
1988 → NOR Flash

1989 → NAND Flash

1995 → MLC NOR

2005 → MLC NAND

**2010 → Intel-Micron 64Gb MLC
NAND in 25nm tech.**

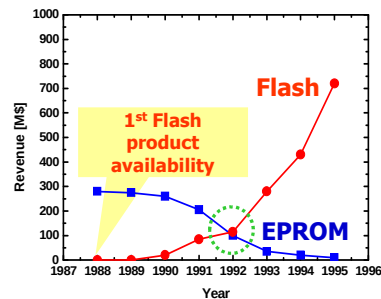


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NVM Market Timeline

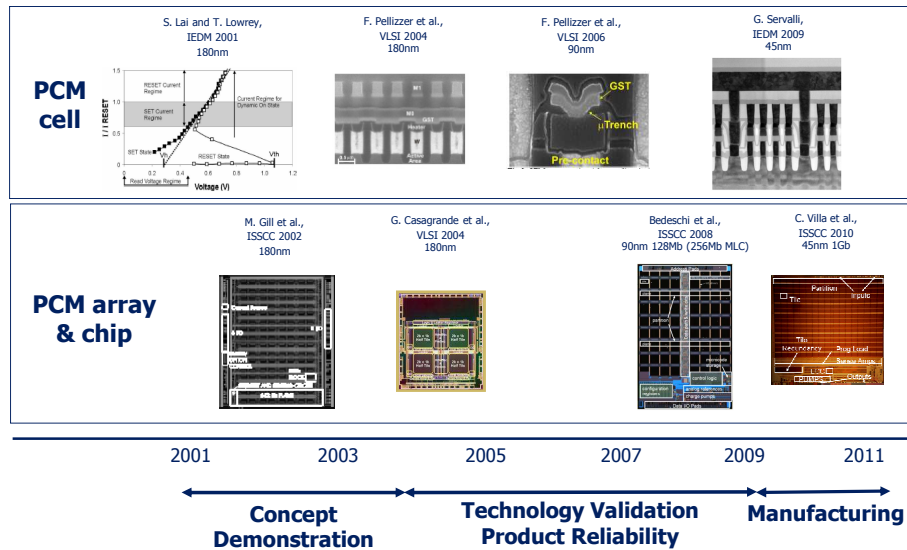
- New memory technologies are rare
 - SRAM, DRAM, EPROM are 30 years old concepts
 - Evolutionary changes for NVM
 - EPROM → E²PROM → Flash
 - Even less innovation for volatile RAM
- Displacement timeline for revenues cross-over
 - 5 years needed from first Flash product availability in 1988 and the revenue crossover with EPROM in 1992

Year of 1 st Shipment	Memory Technology
1969	SRAM
1970	DRAM
1971	EPROM
1988	NOR Flash
1995	NAND Flash
1997	MLC NOR



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History of PCM Development



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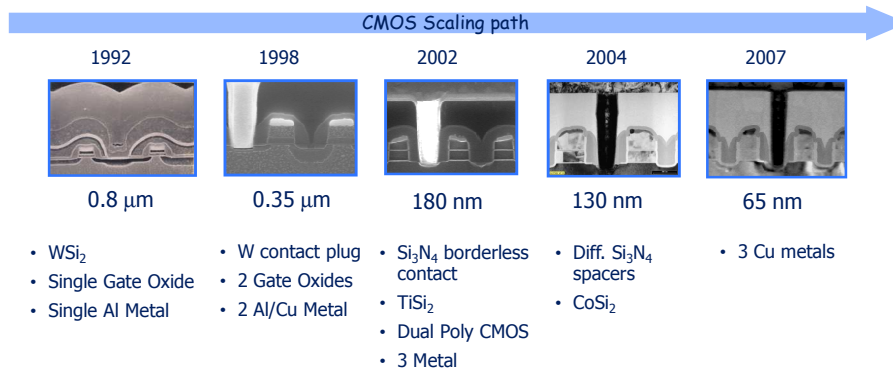
Key Learnings

- Opportunities exist in the NVM market for new memory concepts that can provide a competitive advantage, but...
- ... disruptive innovation takes a lot of time!



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NOR Flash Technology Evolution



Leverage on:

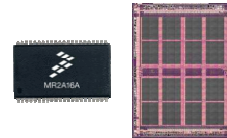
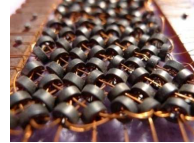
- Standard **CMOS roadmap**
- Specific technologies for Flash



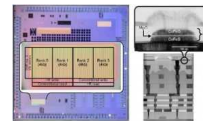
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The MRAM lesson

- Magnetic memory concepts dates back to the 1955, with the introduction of the first magnetic core memory
- In 1995 - Motorola initiates work on MRAM development
- 2003 - A 128 kbit MRAM chip was introduced, manufactured with a 180 nm lithographic process
- 2004 - MRAM becomes a standard product offering at Freescale Semiconductor
- 2006 - Freescale Semiconductor begins marketing a 4-Mbit MRAM chip at 180nm
- **2007 - R&D moving to spin transfer torque MRAM (STT-MRAM)**



Freescale MRAM chip



Fujitsu STT-MRAM chip



Key Learnings

- Opportunities exist in the NVM market for new memory concepts that can provide a competitive advantage, but...
- ... disruptive innovation takes a lot of time!
- To be successful in the memory market, a new NVM concept must be scalable well beyond the actual leading-edge technology node



Chalcogenide Alloys

Alloys with an element of the VI group of the periodic table, usually combined with IV and V group elements

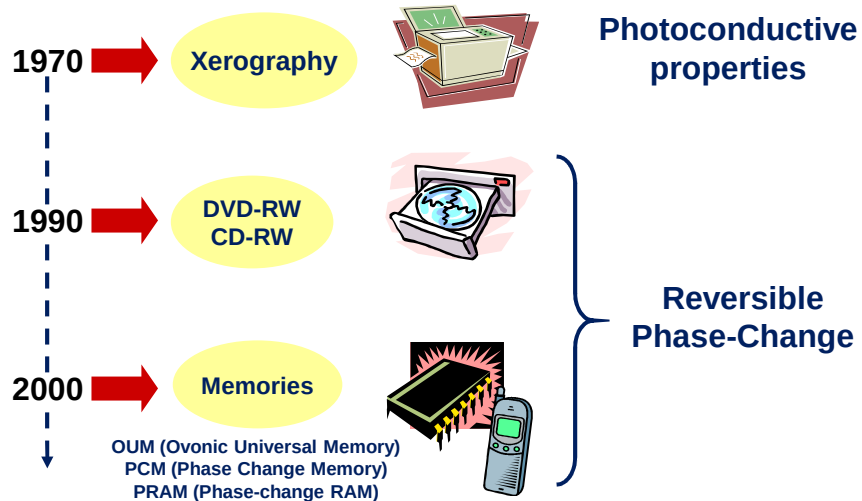
IVA	VA	VIA	VIIA		
C	N	O	F	• As ₂ S ₃	• As ₂ Se ₃
Si	P	S	Cl	• As ₂ Te ₃	• Sb ₂ Te ₃
Ge	As	Se	Br	• SnSb ₂ Te ₄	• Ge ₄₁ Sb ₁₂ Te ₄₁ Se ₆
Sn	Sb	Te	I	• GeTe	• Ge ₂ Sb ₂ Te ₅ (GST)
Pb	Bi	Po	At	• Ge ₁ Sb ₄ Te ₇	

↑
Chalcogenic elements



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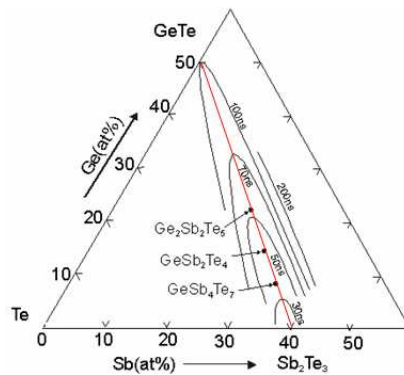
Chalcogenide Applications



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Phase Change Memory: New Materials

Certain alloys containing one or more group VI elements (Chalcogenides) exhibit reversible transition between the disordered and ordered atomic structure



Pseudo-binary $(\text{GeTe})_x$ - $(\text{Sb}_2\text{Te}_3)_y$ compositions:

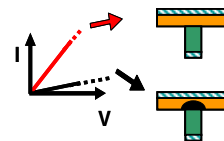
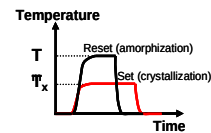
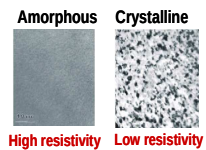
- GeSb_4Te_7
- GeSb_2Te_4
- $\text{Ge}_2\text{Sb}_2\text{Te}_5$



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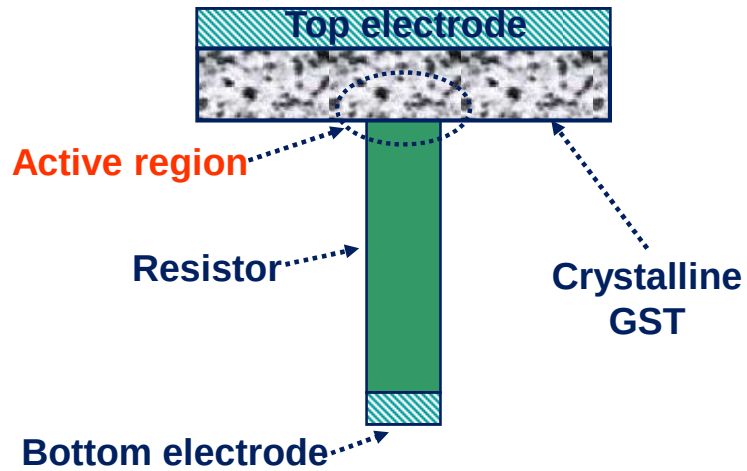
Phase Change Memory (PCM)

- Storing mechanism
 - Amorphous/poly-crystal phase of chalcogenide alloy ($\text{Ge}_2\text{Sb}_2\text{Te}_5$ – GST)
- Writing mechanism
 - Current-induced Joule effect
- Sensing mechanism
 - Resistance change of the GST
- Cell structure
 - 1 transistor, 1 resistor (1T/1R)



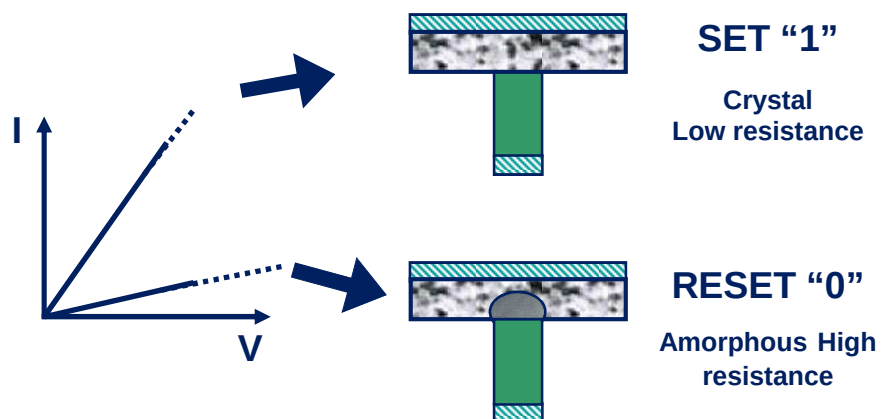
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PCM Storage Element



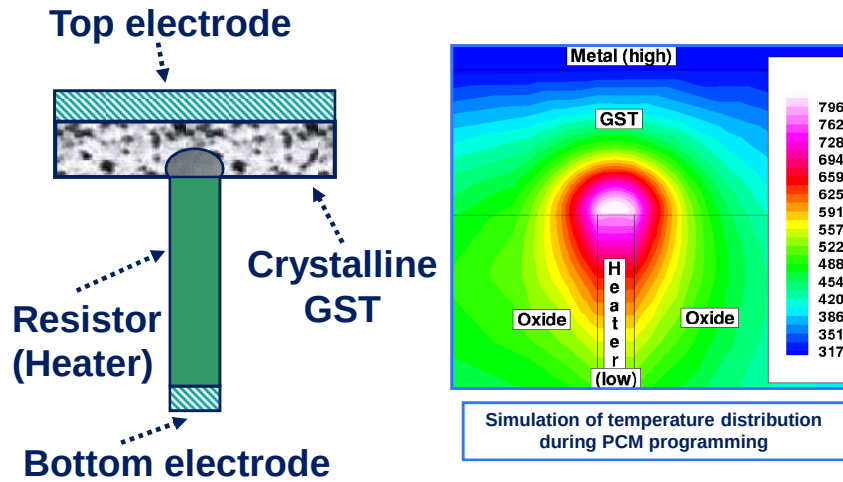
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Sensing Mechanism



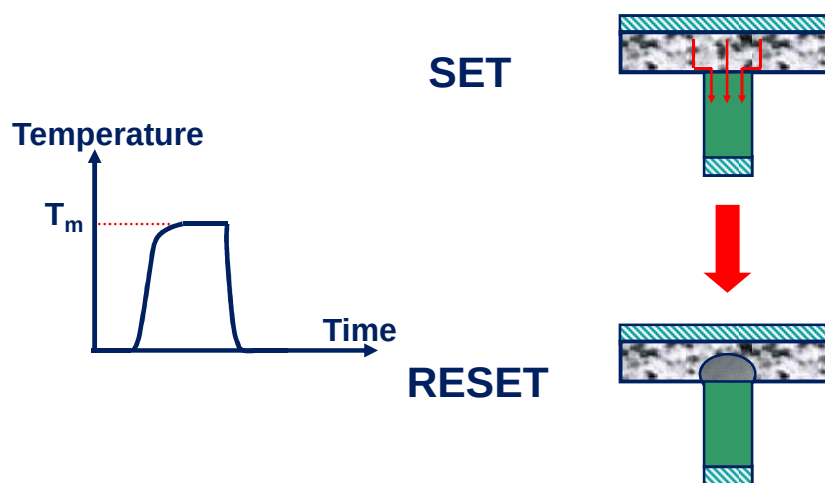
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Joule Heating



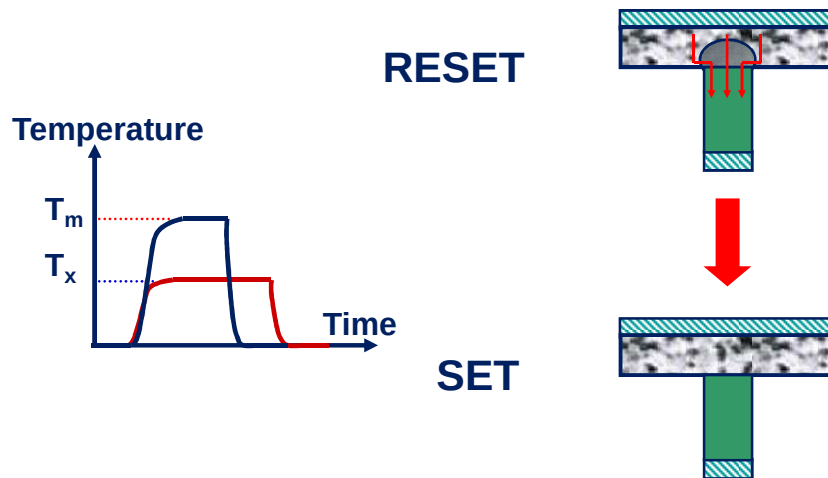
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PCM Operating Principles - Set to Reset



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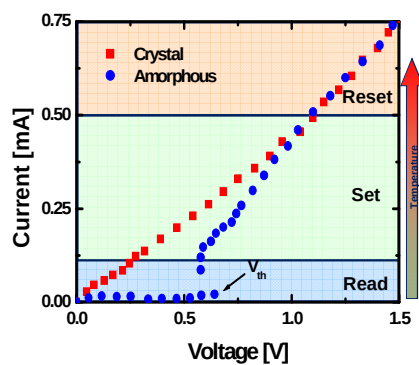
PCM Operating Principles - Reset to Set



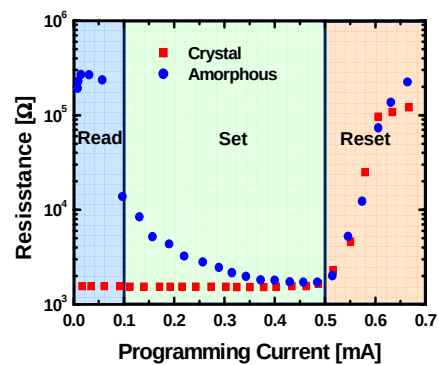
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Fundamental Characteristics

Current-voltage curve

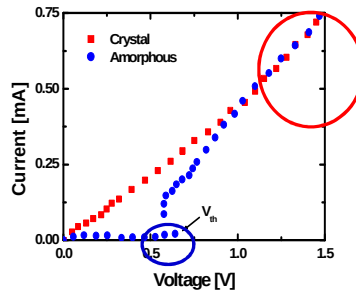


Programming curve

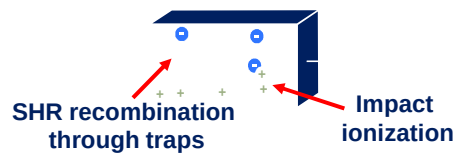


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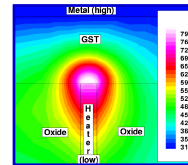
Basic Physical Mechanisms



Electronic Switching (Reversible)

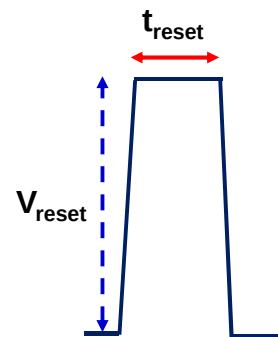
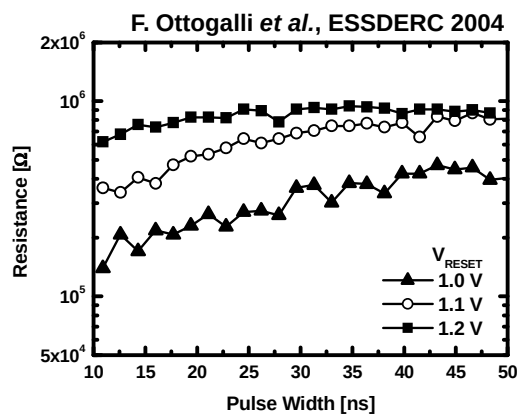


Joule Heating and Phase Change (Memory Effect)



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Reset Programming



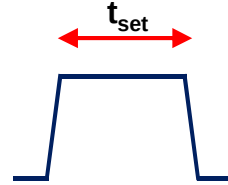
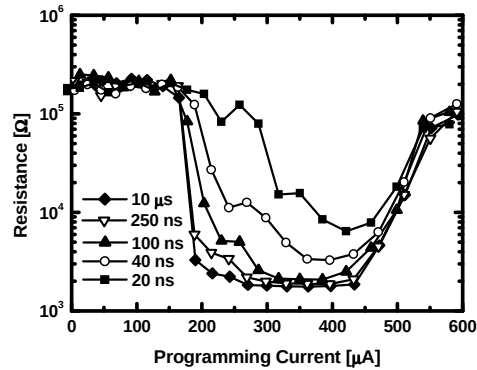
PCM cell can be reset in 10ns with a good reading window



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Set Pulse Width

F. Ottogalli *et al.*, ESSDERC 2004



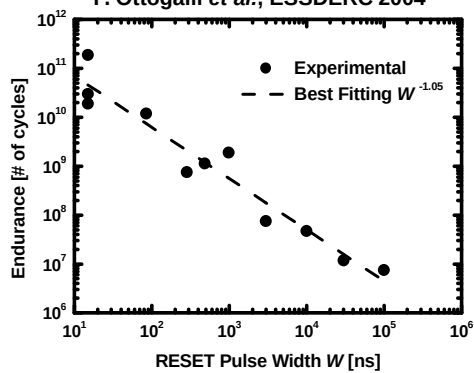
PCM cell can be set with 20ns pulses
still maintaining a good reading window



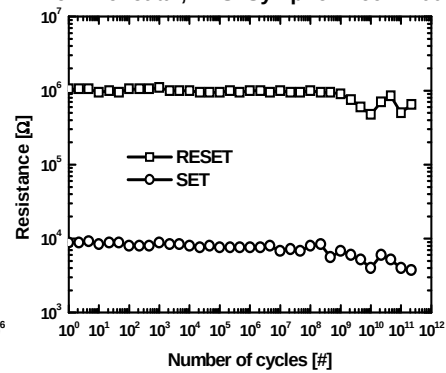
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Endurance

F. Ottogalli *et al.*, ESSDERC 2004



F. Pellizzer *et al.*, VLSI Symp. on Tech. 2003



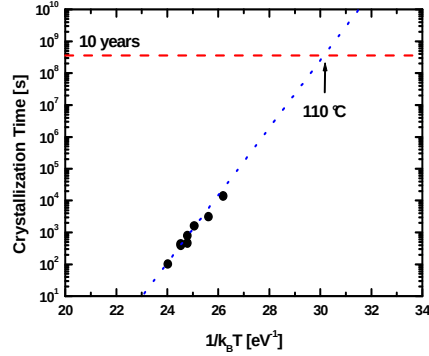
- Reset operation has the strongest impact on endurance performance
- More than 10^{11} cycles have been demonstrated



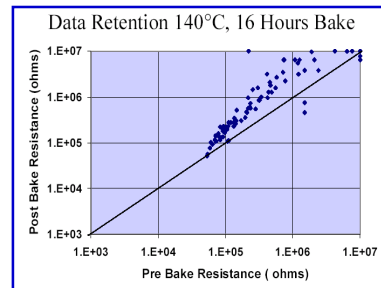
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Data Retention

A. Pirovano *et al.*, IEDM Tech. Dig. 2003



M. Gill *et al.*, ISSCC 2002



- The activation energy is 2.6 eV
- 10 years at 110° C have been extrapolated

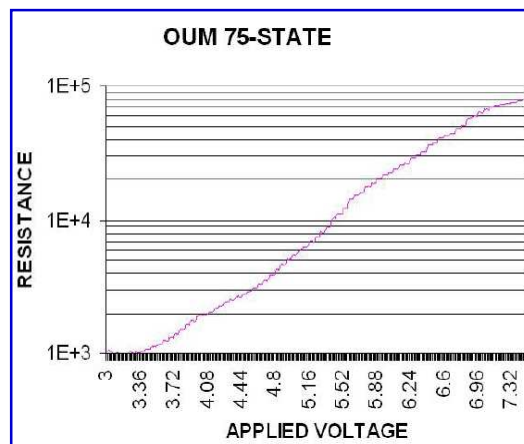


More than 300
years at 85° C



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Multilevel Capabilities



S. Lai, IEDM Tech. Dig. 2003



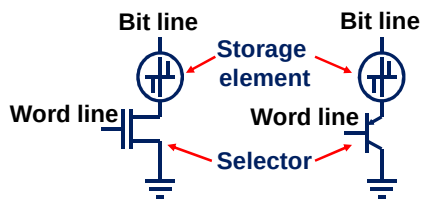
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PCM Cell Structure

Cell Structure: 1 selector + 1 storage element

Transistor

- BJT
- MOSFET
- Diode



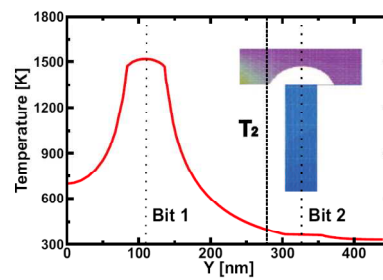
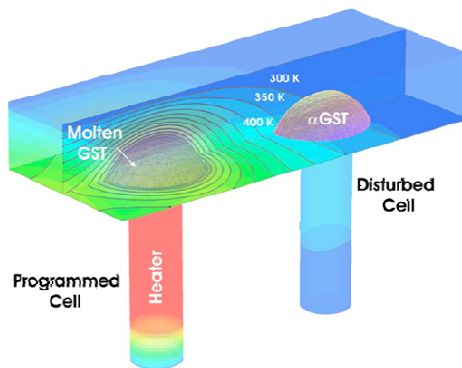
Resistor: heater/material

- Heater
 - Sub-litho contact
 - μ Trench
 - Planar options
- Material
 - GST



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Program disturb issue

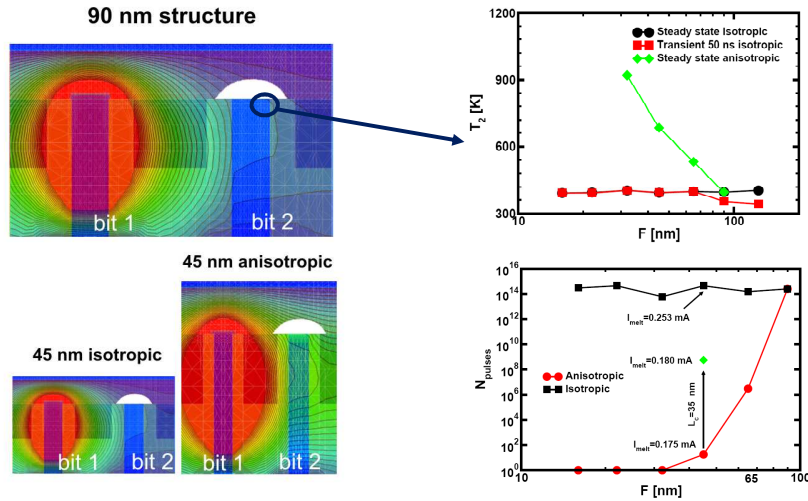


The program operation induces unwanted heating on adjacent bits that may results in thermal disturb

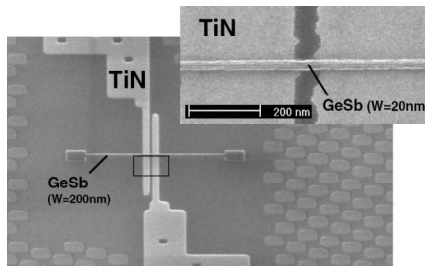


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Cross-talk results

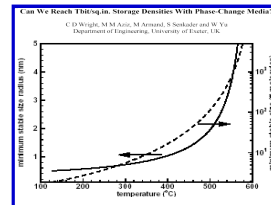


Ultimate scalability of PCM

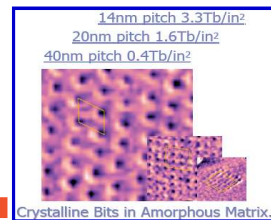


Device functionality demonstrated on 60 nm² active area

Phase change mechanism appears scalable to at least ~5nm



D. Wright *et al.*, EPCOS 2004

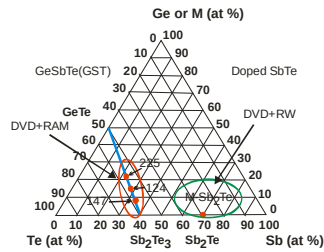


C. Lam, SRC NVM Forum 2004

Materials for Phase Change Memory

Many chalcogenide materials are available for use in solid state memories, exploiting the experience of optical disk research

- But other requirements must be satisfied:



- Electronic switching capability with reasonable switching voltage
- Sufficiently low set resistance for reading performances
- Sufficiently low melting temperature for program performances
- Stability under million of cycles
- Higher crystallization temperature for better retention

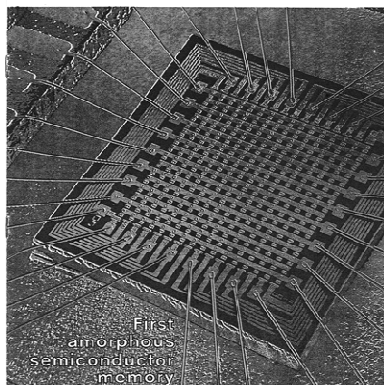
From optical disk experience Ge, Sb, Te, In, Si compounds are most suitable materials for employment in solid state devices



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PCM yesterday

Electronics



1970

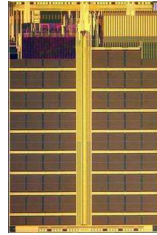
Die: 122 mil X 131 mil
Capacity: 256 bits
Reset: ~200 mA, < 25V, 5 μ s
Set: 5 mA, ~ 25V, 10 ms
Read: 2.5 mA, < 5V

"Nonvolatile and Reprogrammable, the Read-Mostly Memory is Here,"
 R. G. Neale, D. L. Nelson, and
 Gordon E. Moore, *Electronics*
 (Sept. 1970) p. 56.



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90nm Technology - 128Mb "Alverstone" Product



Process Architecture

- PCM cell
 - ▶ Salicided minimum-size BJT selector
 - ▶ Self-aligned "Wall" Structure
 - ▶ 1 Base Contact / 1 Emitter
 - ▶ 0.097 μm^2 cell size
- CMOS architecture
 - ▶ Single gate oxide (8 nm to manage 3V operation)
 - ▶ Dual-flavor poly & CoSi₂
- 3 Cu levels for tight interconnects



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Omneo™ PCM Products Overview

Omneo P5Q PCM

- Single, dual, and quad I/O serial interface
- 128Mb density
- 2.7 – 3.6V supply voltage
- 66MHz clock (50MHz in x4 mode)
- SOIC-16 package



Omneo P8P PCM

- High-performance parallel interface
- 128Mb density
- 2.7 – 3.6V supply voltage
- 1.7 – 3.6V i/o voltage
- 0.7 MB/s programming time
- 56l TSOP; 64b Easy BGA package options

Omneo PCM Delivers Value to Embedded Applications

- Byte alterable, No erase required, Over-write capability
- 1M write cycles delivers 10X flash endurance capability



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PCM: Advantages and Issues

- Main advantages
 - Fast write (<100ns)
 - Good read signal window (factor ten in resistance)
 - Medium/low voltage write
 - Long endurance
 - Cell size comparable to Flash and DRAM
 - Good scalability
 - MLC capabilities
- Main issues
 - Process integration for GST
 - Heater-GST interface optimization
 - Writing current reduction
 - Retention at very high temperature (150° C)

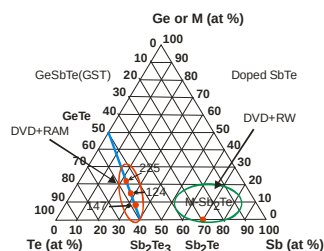


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PCM Active Material

Despite $\text{Ge}_2\text{Sb}_2\text{Te}_5$ has been demonstrated a good material for PCM fabrication, many other chalcogenide materials are available for use in solid state memories, exploiting the experience of optical disk research

But other requirements must be satisfied:



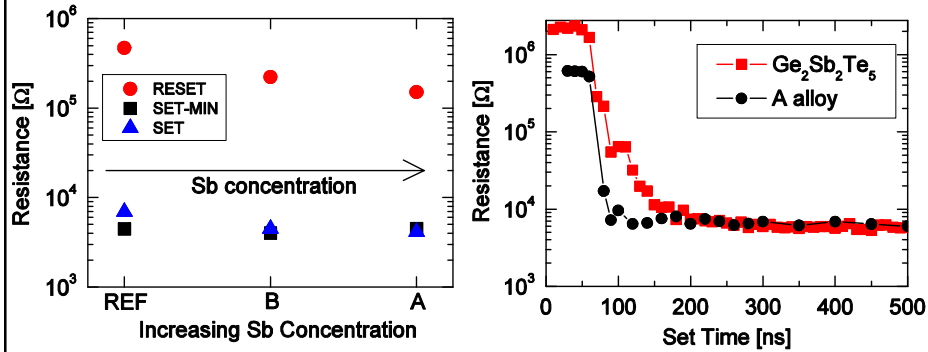
- Electronic switching capability with reasonable switching voltage
- Sufficiently low set resistance for reading performances
- Sufficiently low melting temperature for program performances
- Stability under million of cycles
- Higher crystallization temperature for better retention

From optical disk experience Ge, Sb, Te, In, Si compounds are most suitable materials for employment in solid state devices



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Fast Crystallization Alloy



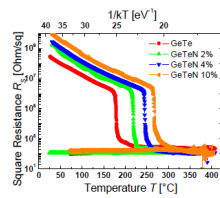
M. Boniardi et al., IMW 2010

- Decrease of the reset resistance with the increase in the Sb concentration
- Convergence of the set level to the minimum set
- Faster crystallization

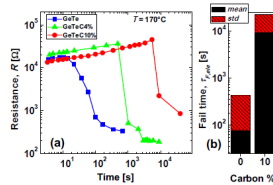


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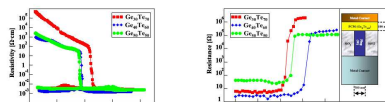
Higher-Temperature Alloy



"N-doped GeTe as Performance Booster for Embedded Phase-Change Memories"
A. Fantini et al., IEDM 2010



"On Carbon doping to improve GeTe-based Phase-Change Memory data retention at high temperature"
G. Betti Beneventi et al., IMW 2010



"Electrical Performances of Tellurium-rich $\text{Ge}_x\text{Te}_{1-x}$ Phase Change Memory"
G. Navarro et al., IMW 2011

Fig. 1. Resistivity measurements on 100 nm-thick full-sheet wafers (temp. rate: 10°C/min).

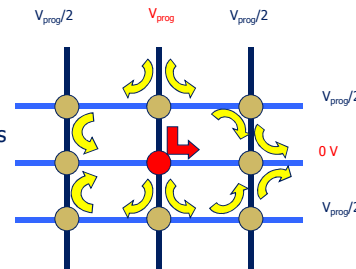
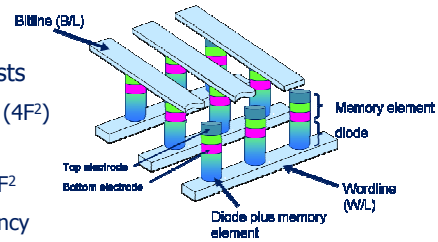
Fig. 2. First RESET program curves for each microstructure. The resistance of the cells is plotted versus the programming current density. In the inset is described the device structure of the devices.



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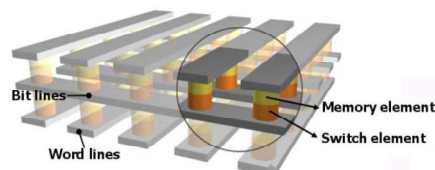
3D Integration Cross-Point Memory

- Crossbar memory attracts great interests
 - “simple” structure and minimum cell size ($4F^2$)
 - low cost
 - suitable for 3D stacking → cell size $(4/n)F^2$
 - array over circuitry → better array efficiency
- The basic cell architecture requires a selector structure to be integrated in the BEOL
 - Parasitic paths exist through neighbouring cells
 - Programming (and also reading) can perturb the array



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A Wide Range of Material Choices



For the selector structure
few concepts have been
proposed so far, all in the
“path finding” phase

Selector device options

- Homojunctions → polySi p/n junctions
- Heterojunctions → p-CuO/n-InZnO
- Schottky diode → Ag/n-ZnO
- Chalcogenide Ovonic Threshold Switching (OTS) materials
- Mixed Ionic Electronic Conduction (MIEC) materials



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Cross-Point Switch Requirements

- Very high forward bias current
 - greater than the switching current
- Low reverse bias current
 - Prevent loss of signal by cross talk
 - Leakage may set the block size
- Composition compatible with memory material
- Low temperature process
- Bipolar operation is preferred

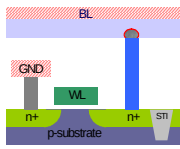
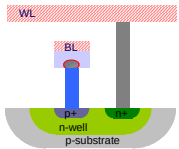
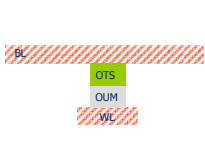


Key Learnings

- Opportunities exist in the NVM market for new memory concepts that can provide a competitive advantage, but...
- ... disruptive innovation takes a lot of time!
- To be successful in the memory market, a new NVM concept must be scalable well beyond the actual leading-edge technology node
- In the last ten years PCM has demonstrated to be able to follow the scaling rules and to have room for entering in the sub-10nm domain



Selectors and PCM Array Architectures

	MOSFET	BJT/Diode	OTS
Process Complexity	No mask overhead for the selector	Dedicated steps for the p-n-p junction integration	Dedicated steps in the BEOL
Cell Size	Larger ($\sim 20F^2$)	Smaller ($\sim 5F^2$)	3D cross-point ($\sim 4F^2/n$)
Memory Array Organization	Conventional	Innovative	Ground-breaking
Application	Embedded memory	High density/ High Performance	Very high density
Schematic Cell Structure Cross-section			



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Embedded PCM (ePCM)

90nm e-NVM Benchmark

	EEPROM	P-FLASH	ePCM
Cell size (λ^2)	111	27	36
Erase block	byte	page	bit
Erase time (Set)	1.5ms	2.5ms	300ns
Program time (Reset)	1.5ms	5μs	100ns
Program voltage	15V	±10V	4V
Program current / bit	< 1μA	50μA	400μA
Endurance	500K	100K	1M
Retention	150C	150C	85C

- For high granularity e-NVM applications e-PCM is the best solution in terms of P/E energy per page and P/E throughput

e-PCM competitive advantage



- Easier to be integrated with advanced logic:
 - lower voltage
 - minor impact on CMOS front-end
- Better performance:
 - single bit over-write
 - programming speed
 - endurance
- Cost-effective:
 - less additional masks (only 3 masks overhead for storage element definition)
 - smaller cell vs. EEPROM, smaller over-head vs. flash

IMW 2010

STMicroelectronics

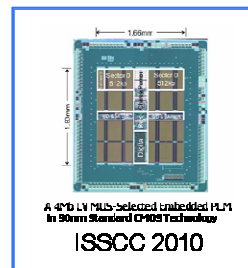
Phase Change Memory Technology for Embedded Non Volatile Memory Applications for 90nm and Beyond

R. Annunziata¹, P. Zuhni², M. Borghesani², G. De Santis¹, L. Scotti¹, C. Prelini¹, M. Tosi², I. Tortorelli² and F. Pellicani²

¹ STMicroelectronics, Via C. Olivetti 2, 20041 Agrate Brianza (Milan), Italy

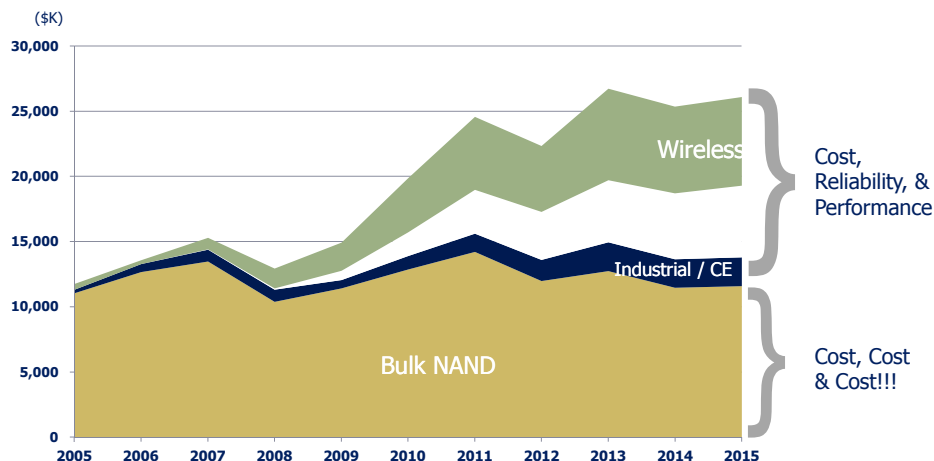
² Numonyx, Via C. Olivetti 2, 20041 Agrate Brianza (Milan), Italy
Tel.: +39-039-6035412, E-mail: roberto.annunziata@st.com

R. Annunziata et al., IEDM 2009



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Stand-Alone NVM TAM Expansion



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Phase Change Memory Key Attributes

- Non Volatility
- Flexibility
 - ▶ No Erase, Bit alterable, Continuous Writing
- Lower power consumption than RAM
- Fast Writes
- Read bandwidth and writing throughput
- eXecution in Place
- Extended endurance

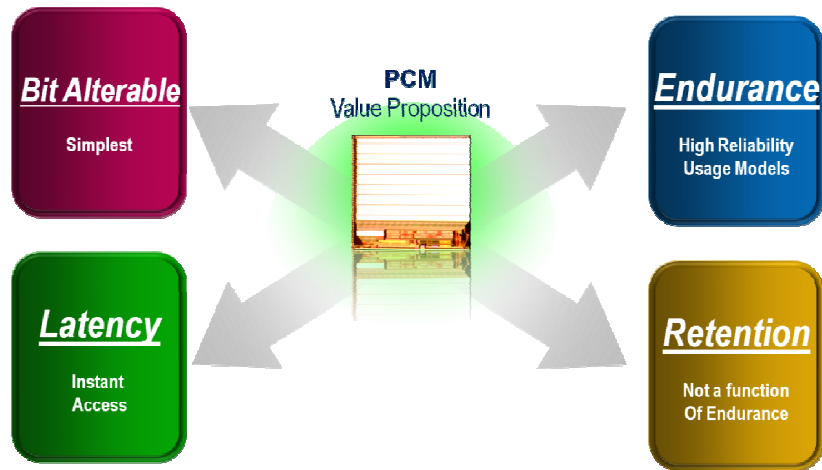
Attributes	PCM	EEPROM	NOR	NAND	DRAM
Non-Volatile	Yes	Yes	Yes	Yes	No
Scaling	sub-2x nm	n/a.	3x nm	2x nm	3x nm
Granularity	Small/Byte	Small/Byte	Large	Large	Small/Byte
Erase	No	No	Yes	Yes	No
Software	Easy	Easy	Moderate	Hard	Easy
Power	~Flash	~Flash	~Flash	~Flash	High
Write Bandwidth	1- 15+ MB/s	13-30 KB/s	0.5-2 MB/s	10+ MB/s	100+ MB/s
Read Latency	50 - 100 ns	200-200 ns	70-100 ns	15 - 50 us	20 - 80 ns
Endurance	10 ¹⁰	10 ⁵ -10 ⁶	10 ⁵	10 ⁴	Unlimited

PCM provides a new set of features combining properties of NVMs with DRAM

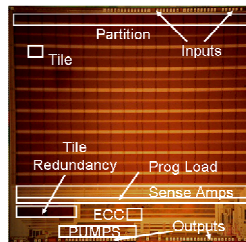


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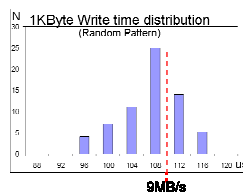
PCM Value Proposition



PCM – NOR Flash legacy



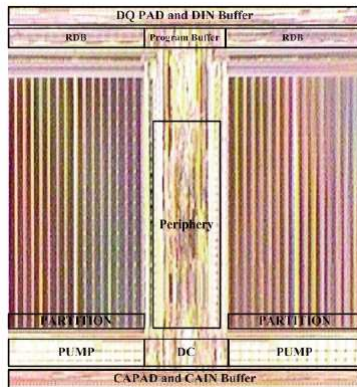
WRITE Performance



C.Villa et al., ISSCC 2010

- Replace NOR in embedded platforms
 - PCM has an edge due to density, scalability, and write speed
- 45nm PCM - 1Gb "Bonelli" specifications
 - NOR Flash legacy spec + bit alterability
 - Chip area: 37.5 mm²
 - Power supply range: 1.7V, 2.0V
 - Temperature range: -40° C, +85° C
- Measured performance
 - Initial access speed: 85ns
 - Max read throughput: 266MB/s
 - Program throughput: 9MB/s

PCM – LPDDR2



H. Chung et al., ISSCC 2011

- Replace (a part of) DRAM
 - PCM has an edge on DRAM due to power and scalability, but it is slower
- 58nm PRAM - 1Gb LPDDR2 specifications
 - LPDDR2 interface
 - Chip area: 63.4 mm²
 - Power supply range: 1.8V, 1.2V
 - Temperature range: -25° C, +85° C
- Measured performance
 - Initial access speed: 76ns
 - IO speed: 800Mbps/pin
 - Program throughput: 6.4MB/s



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PCM Application Opportunities

PCM feature can be exploited by all the memory system, especially the ones resulting from the convergence of consumer, computer and communication electronics

- **Wireless System** to store of XiP, semi-static data and files
 - Bit alterability allows direct-write memory
- **Solid State Storage Subsystem** to store frequently accessed pages and elements easily managed when manipulated in place
 - Caching with PCM will improve performance and reliability
- **Computing Platforms** taking advantage of non-volatility to reduce the power
 - PCM offers endurance and write latency that are compelling for a number of novel solutions

S.Eilert et al., "PCM: a new memory enables new memory usage models", IMW, 2009



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MLC Capability

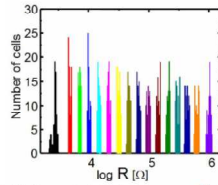


Fig. 11: 10x10 array test structure programmed into 16 levels. Tight, well controlled distributions allow 4bit/cell. Adjustment of pulse slopes depending on resistance is essential to achieve narrow distributions.

"Write Strategies for 2 and 4-bit Multi-Level Phase-Change Memory"
T. Nirschl et al., IEDM 2007

"A Multi-Level-Cell Bipolar Selected Phase Change Memory"
F. Bedeschi et al., ISSCC 2008

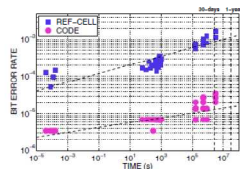
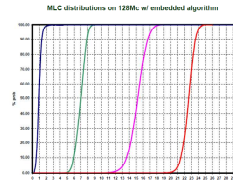


Fig. 7. Bit-error-rate of "reference-cell" and the proposed coding methods on a PCM array of 200 kcells. Measurements have been performed starting at 40 μ s after programming and for extended periods of time.

Distributions placement at 90nm



"Drift-Tolerant Multilevel Phase Change Memory"
N. Papandreou et al., IMW 2011

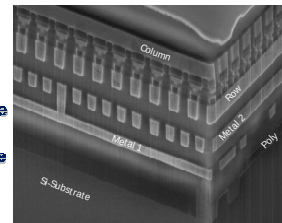
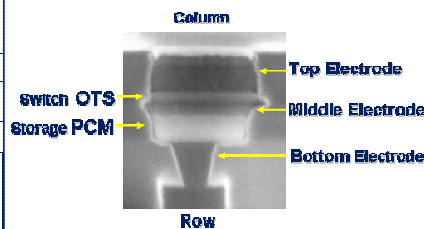


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PCMS Memory Cell Cross-Bar Architecture

Ovonic Threshold Switch, *OTS*, is a two-terminal switch

	OTS
Process Complexity	Dedicated steps in the RFOI
Cell Size	3D cross-point (~4F ² /n)
Memory Array Organization	Ground-breaking
Application	Very high density
Schematic Cell Structure Cross-section	



Intel-Numonyx, IEDM 2009

Chalcogenide materials can be used both for the memory and for the selector (OTS) to form stackable cross point PCM (PCMS)

- True high density cross-bar
- Possible multilayer vertical stacking



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Key Learnings

- Opportunities exist in the NVM market for new memory concepts that can provide a competitive advantage, but...
- ... disruptive innovation takes a lot of time!
- To be successful in the memory market, a new NVM concept must be scalable well beyond the actual leading-edge technology node
- In the last ten years PCM has demonstrated to be able to follow the scaling rules and to have room for entering in the sub-10nm domain
- Replacing an existing memory technology could be a very hard challenge, but the key features of PCM can be effectively exploited for improving existing applications



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